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(54) **WASHING MACHINE**

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(71) Applicant: **SAMSUNG ELECTRONICS CO., LTD.**, Suwon-si (KR)

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(72) Inventors: **Heedong KIM**, Suwon-si (KR); **Taegyu KIM**, Suwon-si (KR); **Minwoo RYU**, Suwon-si (KR); **Jungwook BAE**, Suwon-si (KR); **Euihyun JUNG**, Suwon-si (KR)

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(73) Assignee: **SAMSUNG ELECTRONICS CO., LTD.**, Suwon-si (KR)

(57)

ABSTRACT

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The washing machine according to an embodiment of the disclosure may include a cabinet. The washing machine may include a tub in the cabinet. The washing machine may include a detergent supply device for supplying a detergent to the tub. The washing machine may include a detergent cartridge accommodating the detergent, configured to be drawn in or out the detergent supply device and a cartridge identification member storing information about the detergent accommodated in the detergent cartridge. The washing machine may include a cartridge identification device positioned to, with the detergent cartridge drawn into the detergent supply part, recognize the cartridge identification member.

Related U.S. Application Data

(63) Continuation of application No. PCT/KR2025/000426, filed on Jan. 8, 2025.

Foreign Application Priority Data

Mar. 8, 2024 (KR) 10-2024-0033496

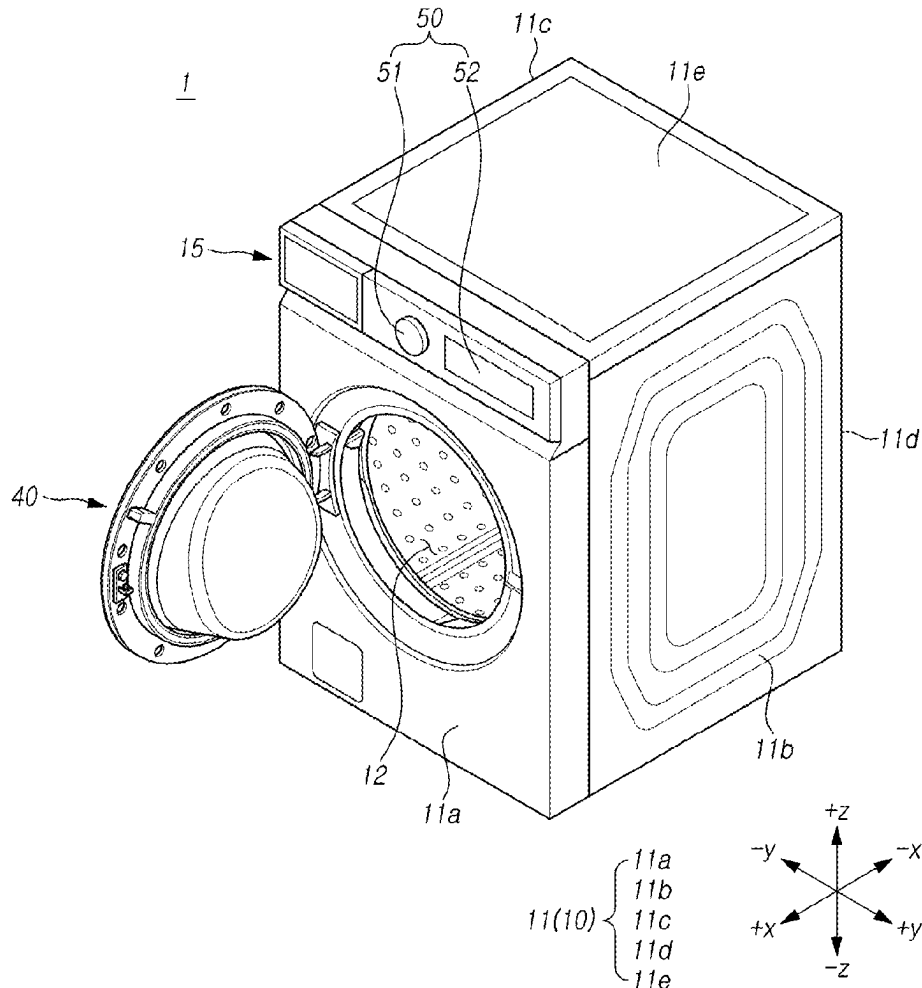


FIG. 1

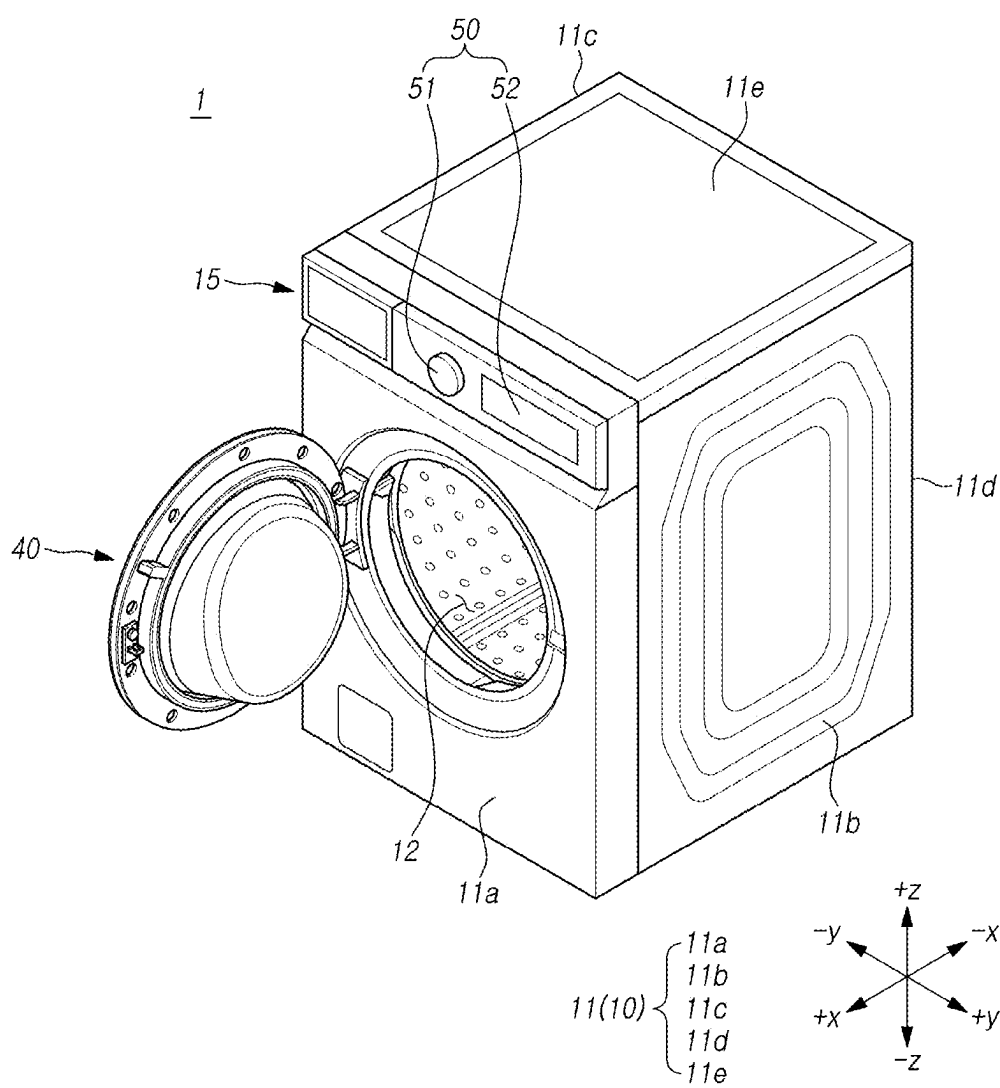


FIG. 2

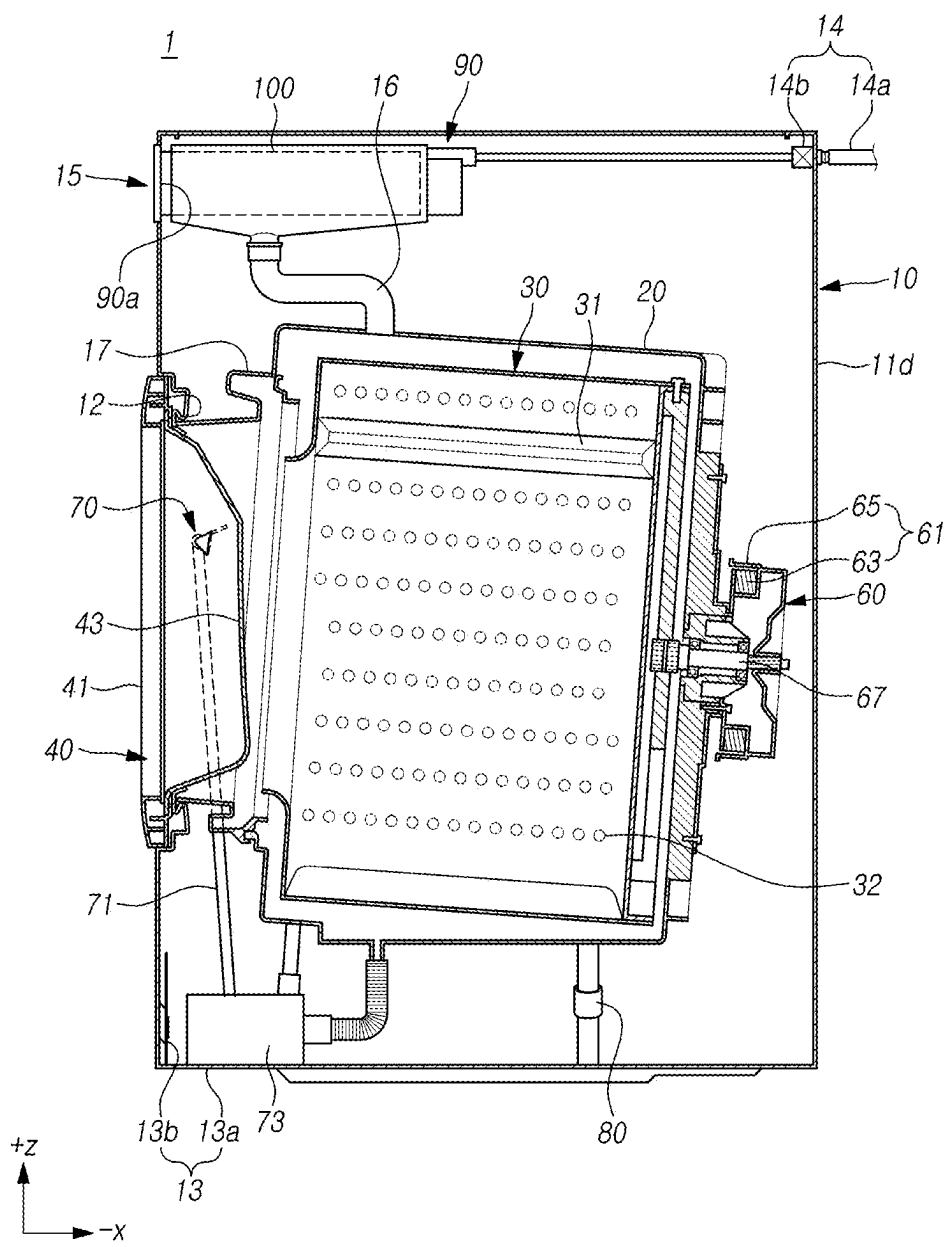


FIG. 3

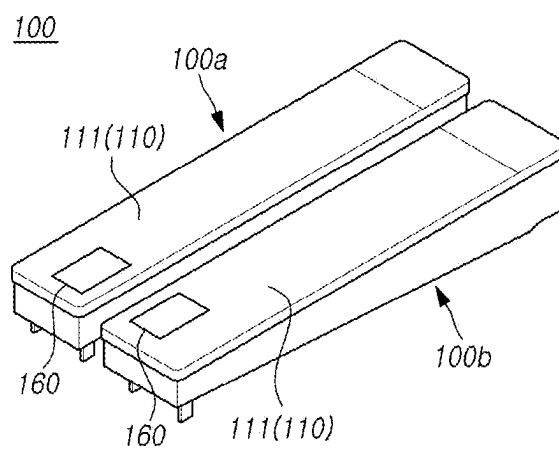


FIG. 4

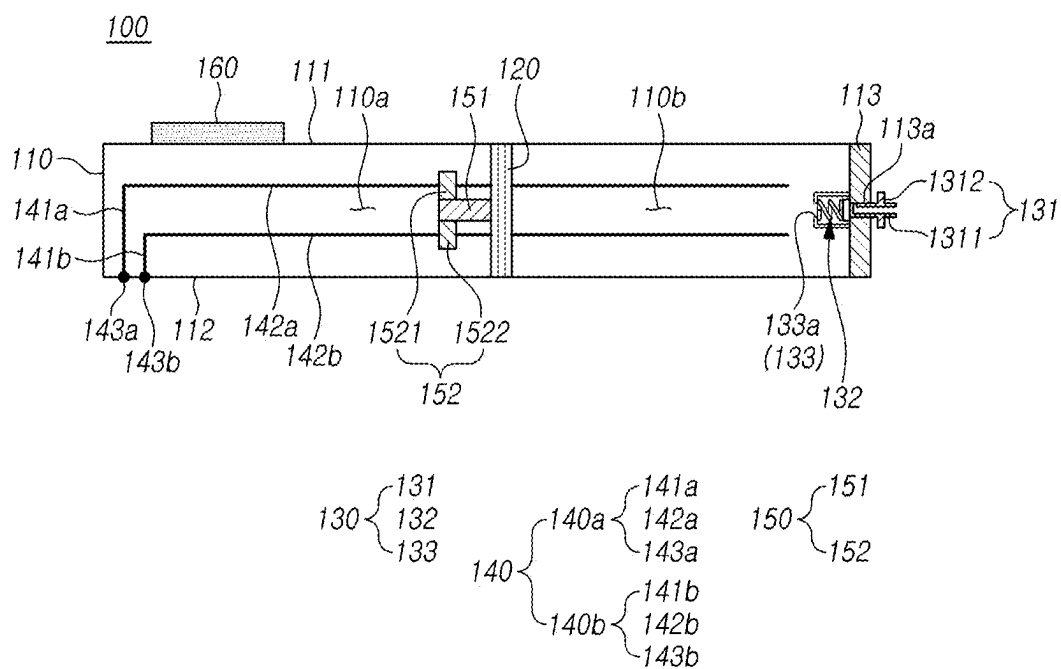


FIG.5

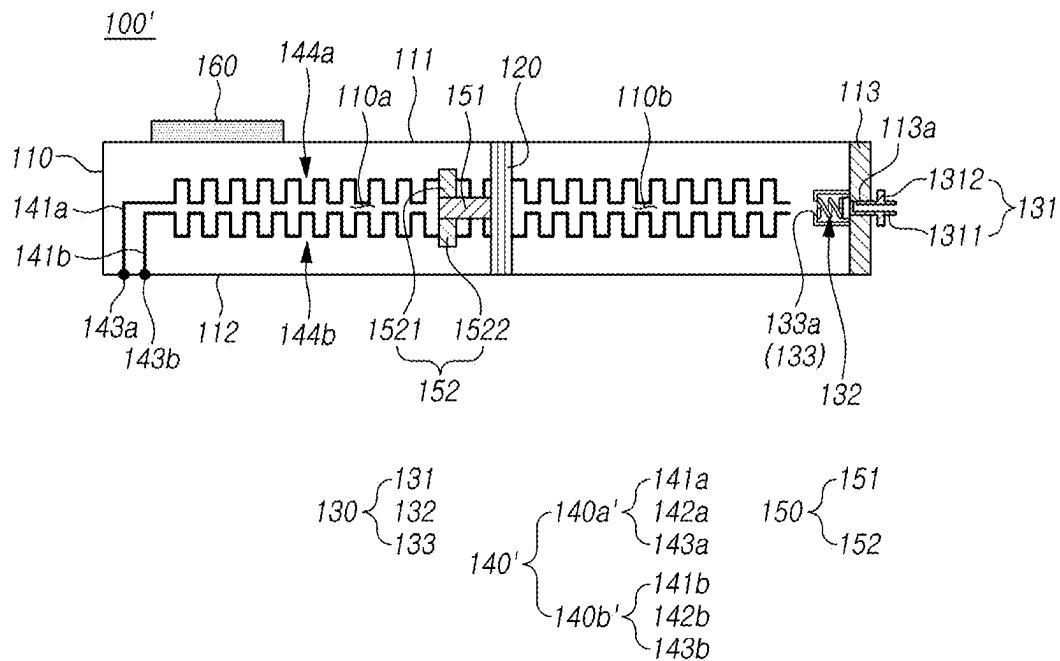


FIG. 6B

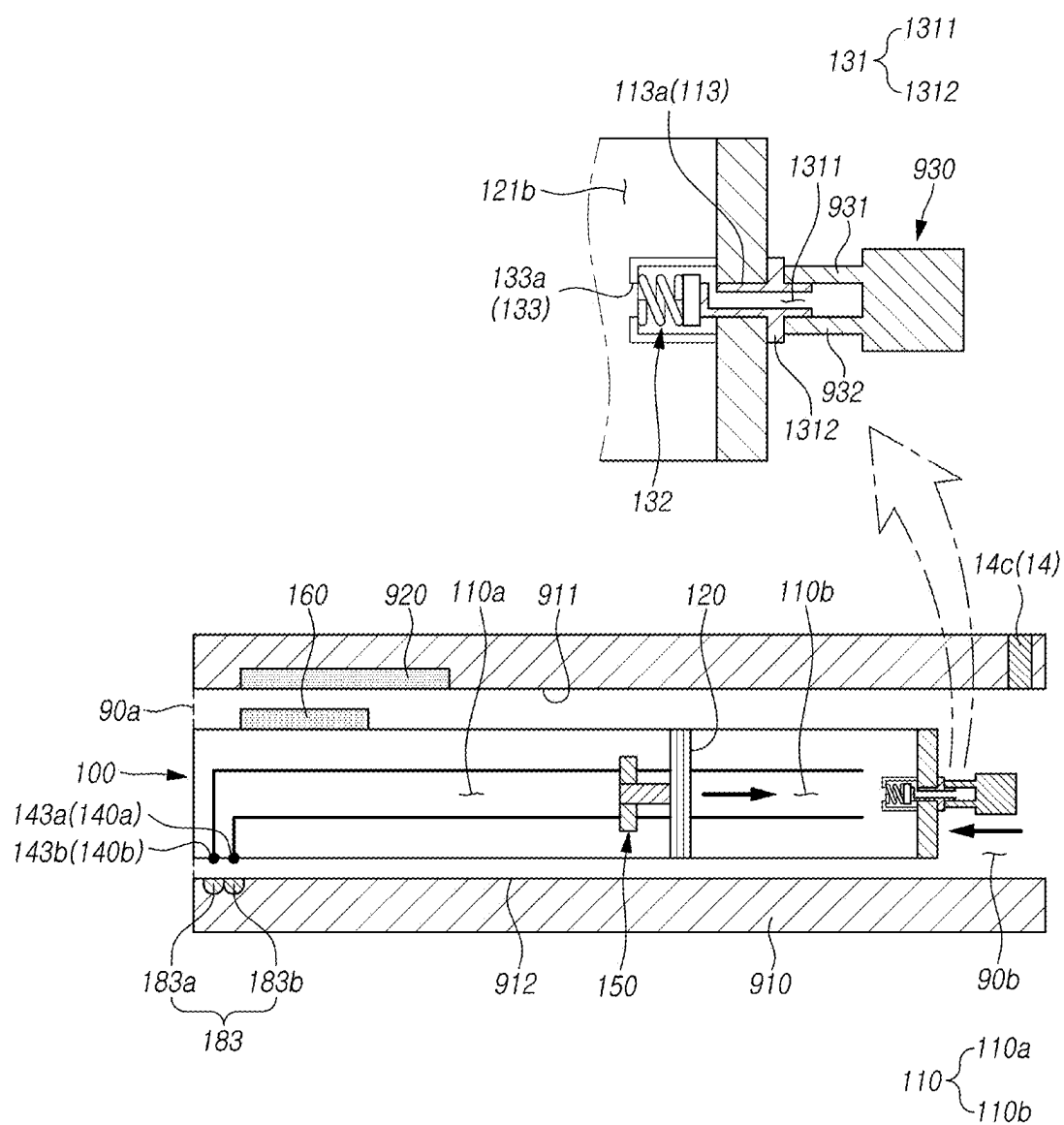


FIG. 7

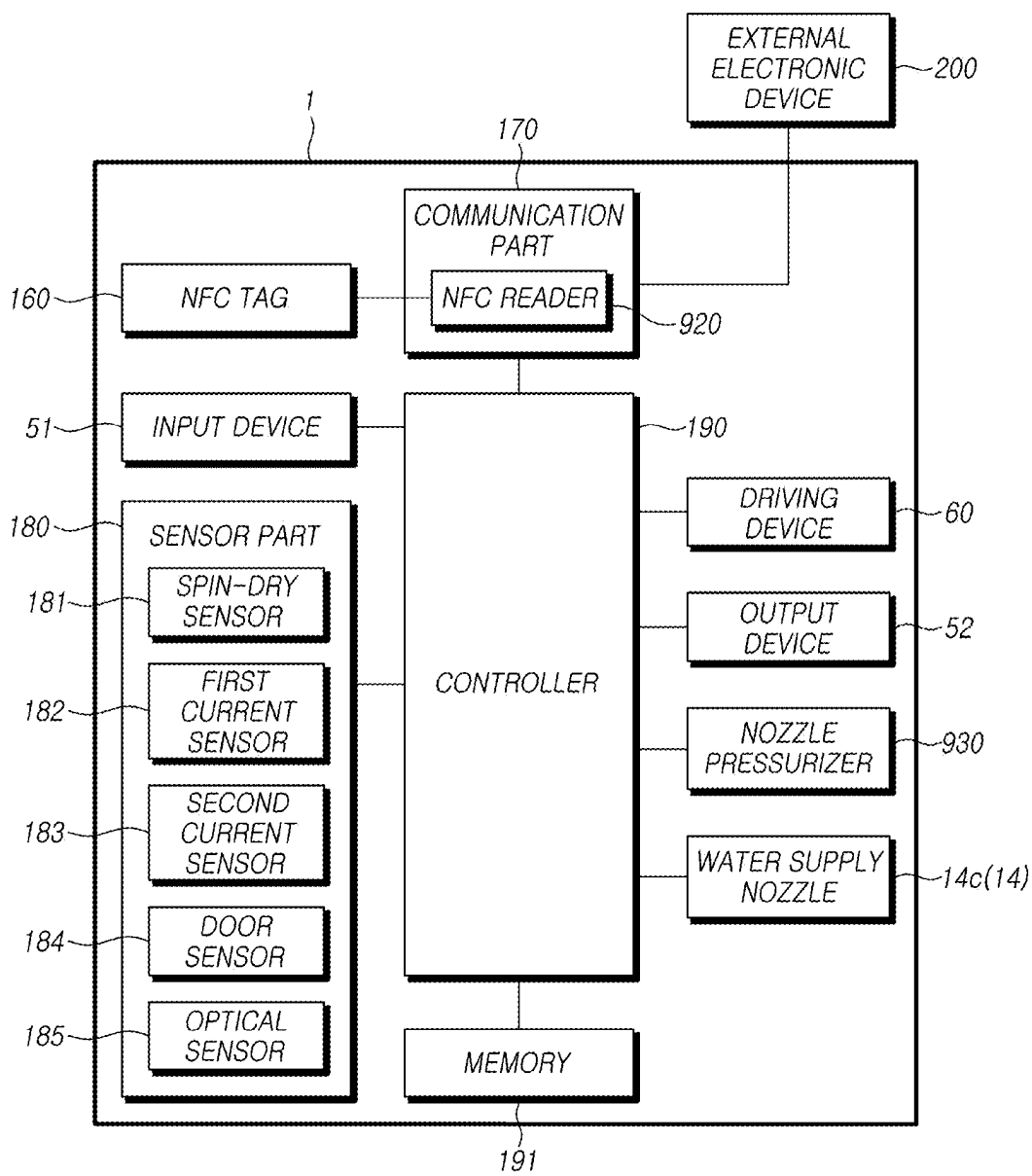


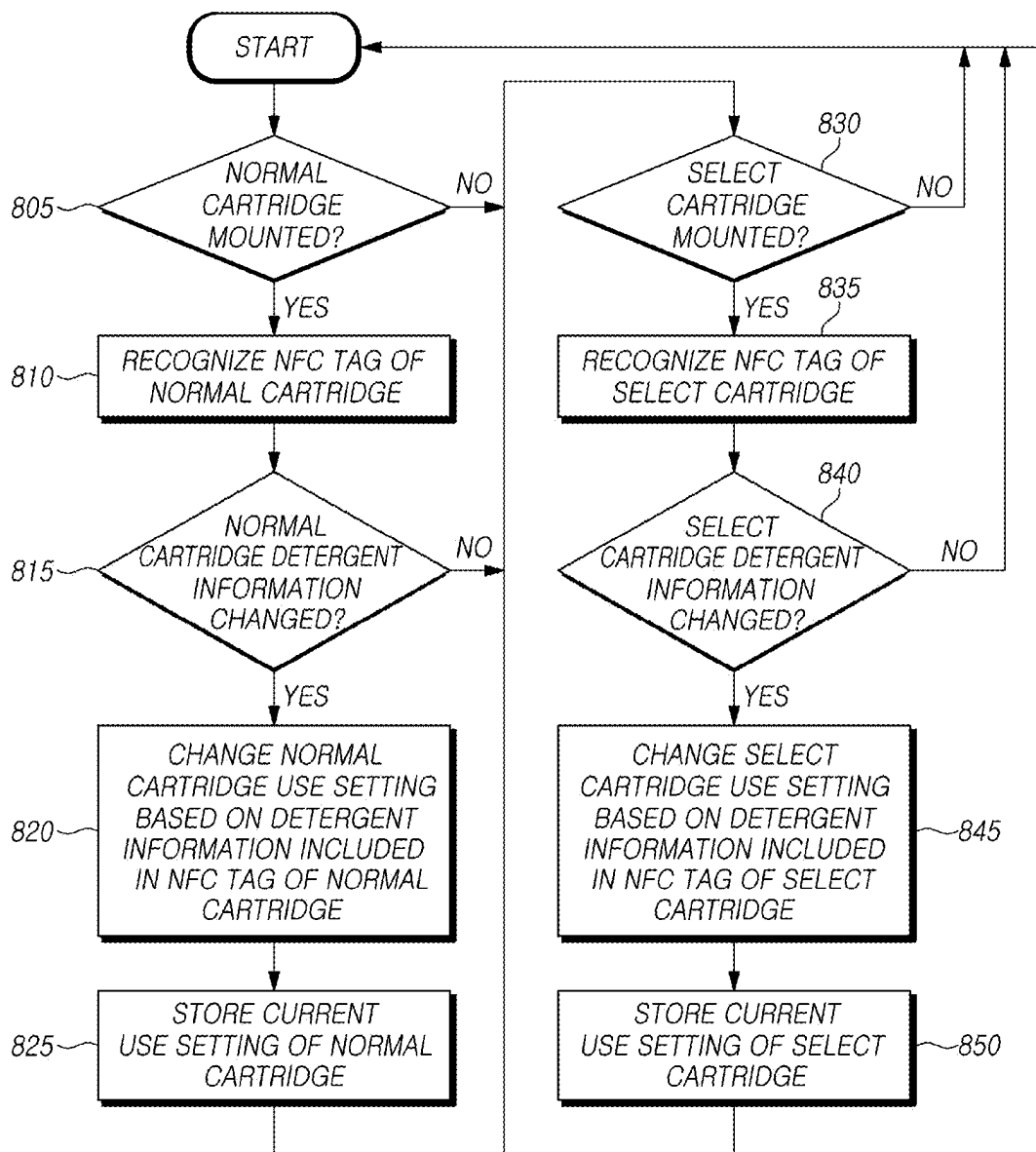
FIG. 8

FIG. 9

CONTENT OF SELECT CARTRIDGE	AUTOMATIC USE SETTING OPERATION OF CARTRIDGE
SOFTENER	1. AUTOMATIC SETTING OF USE SETTING OF SELECT CARTRIDGE TO SOFTENER
NORMAL DETERGENT	1. AUTOMATIC SETTING OF USE SETTING OF SELECT CARTRIDGE TO NORMAL DETERGENT 2. ALARM OF SELECTION OF CARTRIDGE FOR USE IN WASHING 3. ALARM OF IDENTIFYING WHETHER TO USE SOFTENER
NEUTRAL DETERGENT	1. AUTOMATIC SETTING OF USE SETTING OF SELECT CARTRIDGE TO NEUTRAL DETERGENT 2. ALARM OF IDENTIFYING WHETHER TO USE SOFTENER
INFANT DETERGENT	1. AUTOMATIC SETTING OF USE SETTING OF SELECT CARTRIDGE TO INFANT DETERGENT 2. ALARM OF IDENTIFYING WHETHER TO USE SOFTENER

FIG. 10

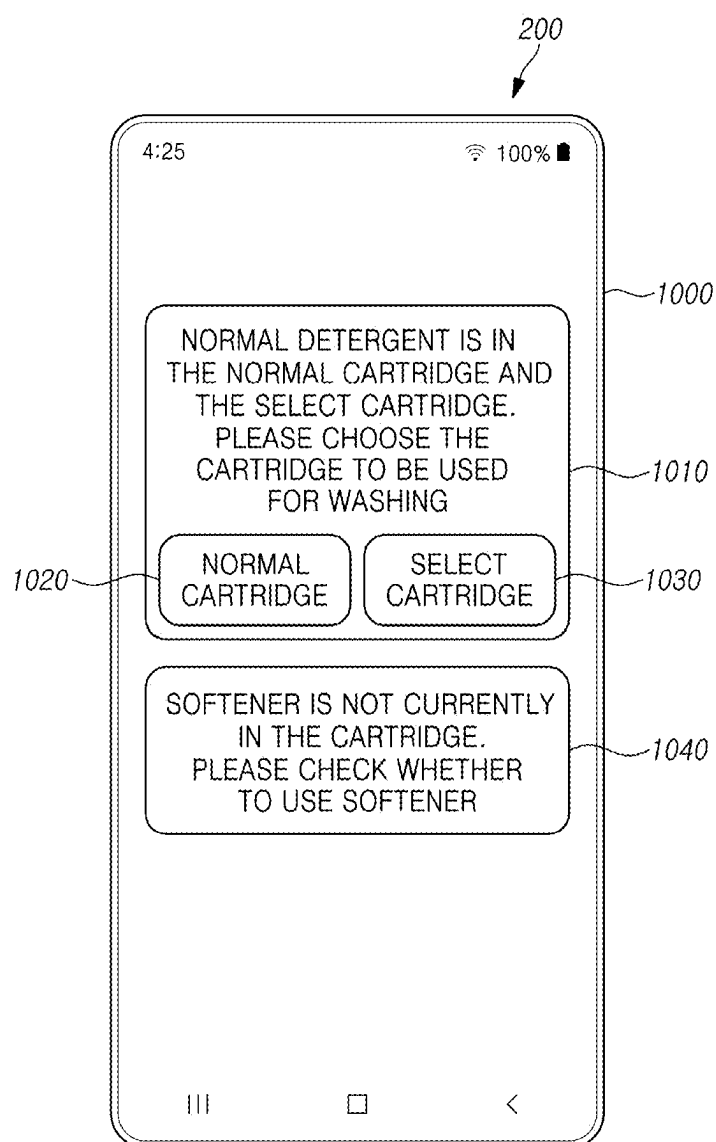


FIG. 11

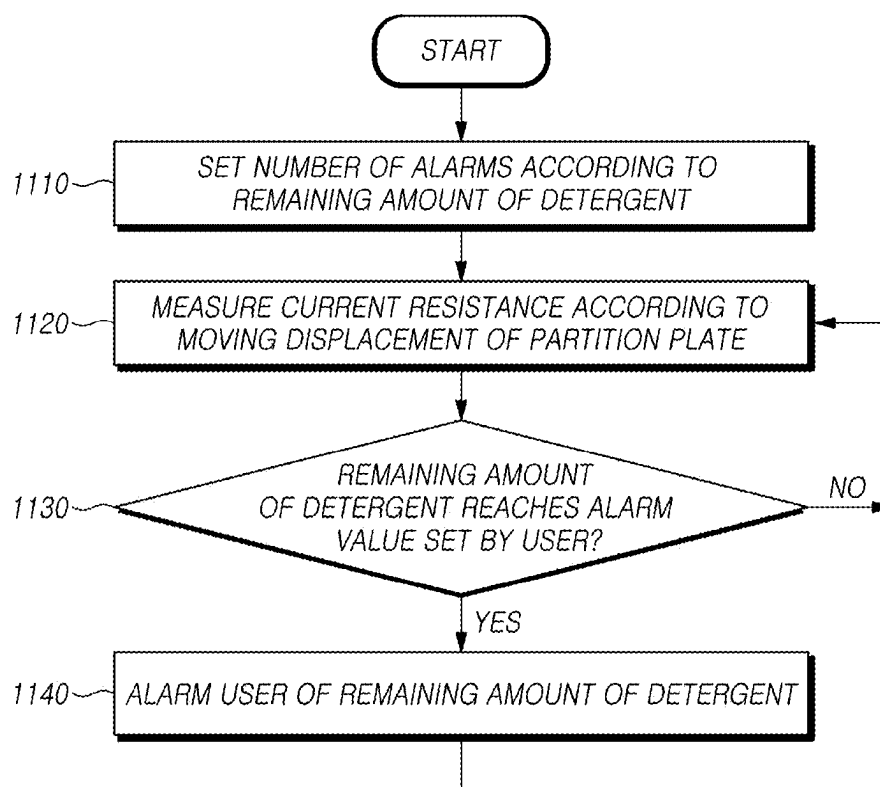


FIG. 12

Alarm Value	Remaining Amount Of Detergent
First Alarm	70% Of Initial Content
Second Alarm	40% Of Initial Content
Third Alarm	20% Of Initial Content
Fourth Alarm	10% Of Initial Content
⋮	⋮
Nth Alarm	xx% Of Initial Content

FIG. 13

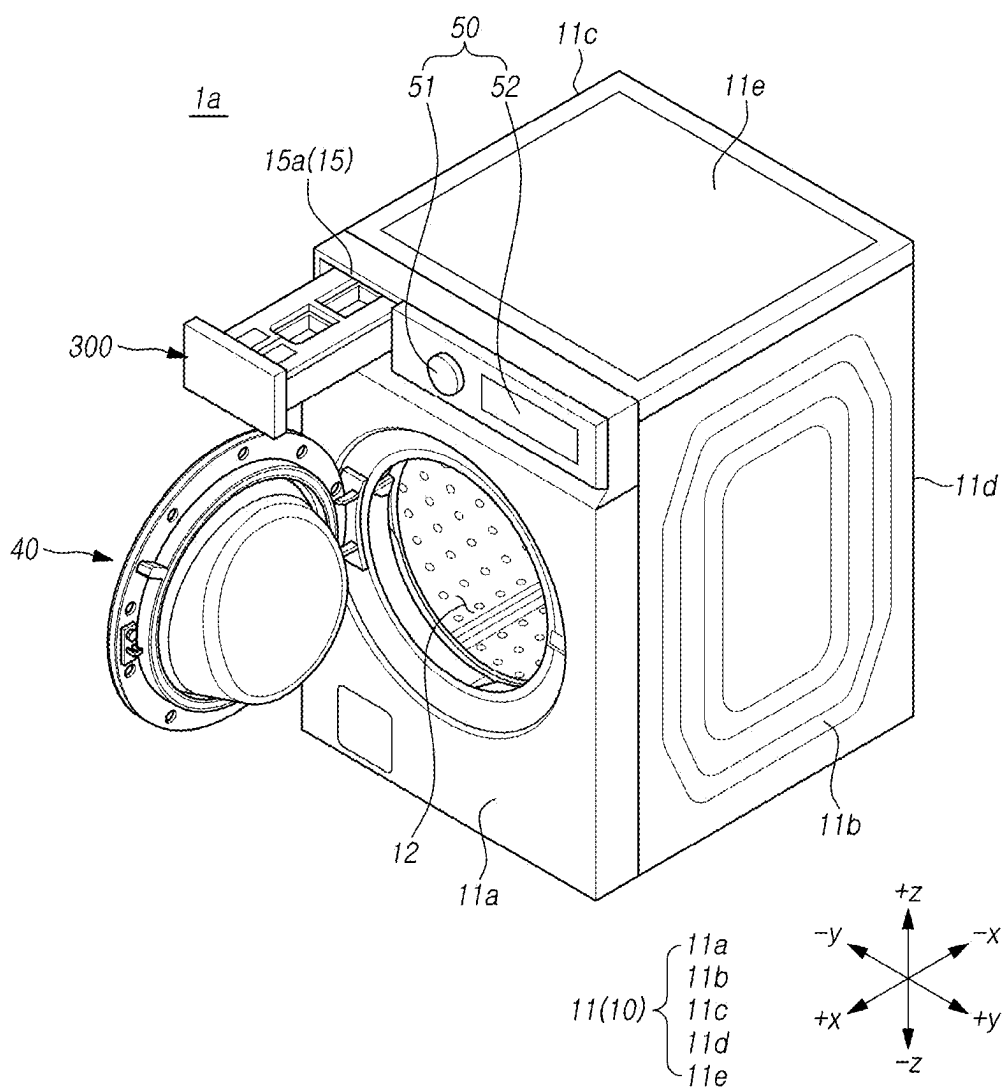


FIG. 14

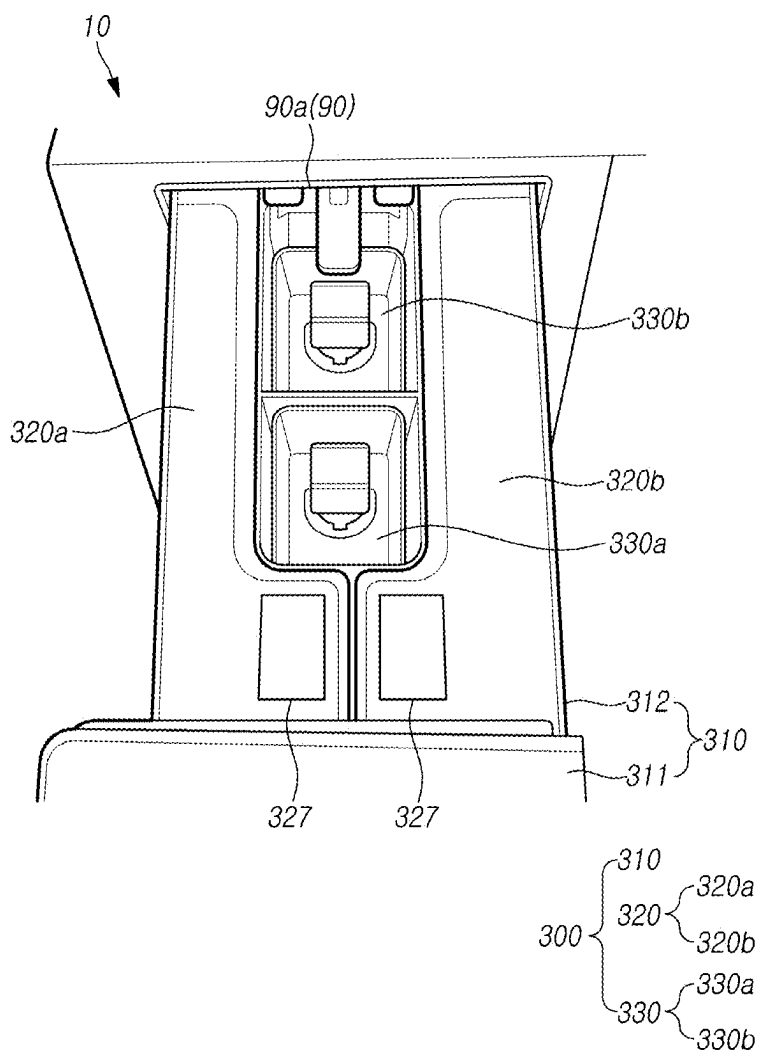
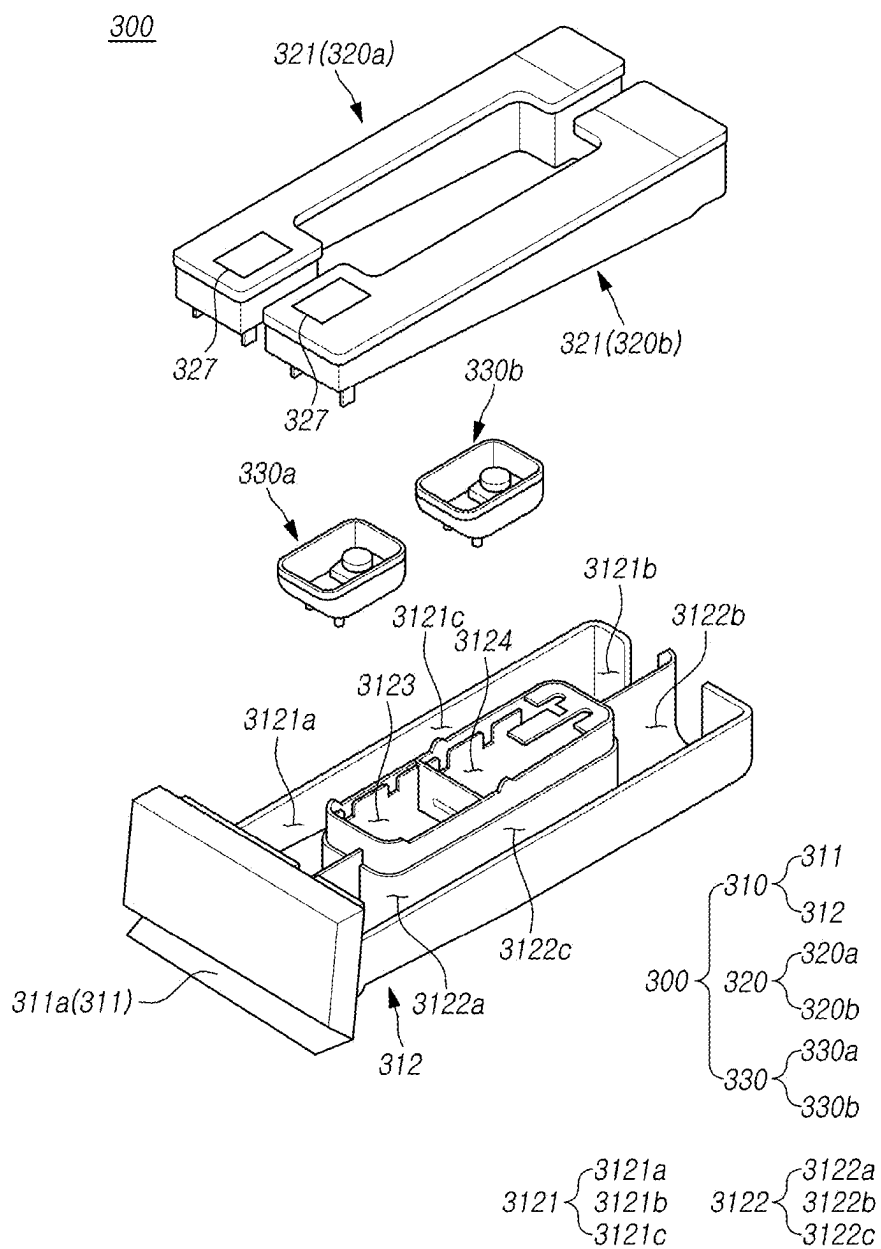


FIG. 15



WASHING MACHINE

CROSS-REFERENCE TO RELATED APPLICATIONS

[0001] This application is a continuation application of International Application No. PCT/KR2025/000426 designating the United States, filed on Jan. 8, 2025, in the Korean Intellectual Property Receiving Office, which claims priority from Korean Patent Application No. 10-2024-0033496, filed on Mar. 8, 2024, in the Korean Intellectual Property Office, the disclosures of which are hereby incorporated by reference herein in their entireties.

BACKGROUND

Field

[0002] Various embodiments of the disclosure relate to a washing machine including a detergent supply device and a method for controlling the washing machine.

Description of Related Art

[0003] In general, a washing machine is a device that stirs laundry, washing water, and detergent put into a washing tub together using a driving force of a motor so that washing is performed through friction between the laundry, washing water, and detergent.

[0004] The washing machine includes a detergent supply device for automatically or manually supplying a detergent so that the detergent is evenly mixed with water supplied while water is supplied into the tub.

[0005] The detergent supply device includes several detergent containers depending on the type of detergent accommodated and the automatic/manual supply function of the detergent. For example, the detergent supply device includes an automatic detergent container for automatic supply of detergent and a manual detergent container for manual supply of detergent.

[0006] Typically, information about the detergent such as the type of the detergent accommodated in the automatic detergent container, the concentration of the detergent, and the amount of detergent put in during washing (e.g., detergent input amount) is manually input to the washing machine by the user. The washing machine supplies the detergent into the tub during washing based on the input detergent information.

[0007] For example, when the user intends to use the automatic detergent container, the user should make a manual configuration on the use settings of the automatic detergent container based on the information about the detergent accommodated in the automatic detergent container through the control panel of the washing machine or an external electronic device.

[0008] Further, when changing the type of the detergent accommodated in the automatic detergent container, the user is required to change the use settings of the automatic detergent container according to the changed detergent.

[0009] In other words, the user has the inconvenience of being required to make a manual configuration on the use settings of the automatic detergent container according to the detergent accommodated in the automatic detergent container or manually change the use settings when changing the detergent accommodated in the automatic detergent container.

SUMMARY

[0010] Various embodiments of the disclosure may provide a washing machine that has a detergent cartridge provided with an NFC tag, and obtains information about the detergent cartridge and the detergent accommodated in the detergent cartridge in an NFC manner and automatically configures use settings of the detergent cartridge based on the obtained information.

[0011] Various embodiments of the disclosure may provide a washing machine that detects the remaining amount of detergent in the detergent cartridge by measuring the variable resistance that changes according to the amount of detergent used.

[0012] A washing machine according to an embodiment of the disclosure may include a cabinet. The washing machine may include a tub disposed in the cabinet. The washing machine may include a detergent supply part for supplying a detergent to the tub. The washing machine may include a detergent cartridge accommodating the detergent, configured to be drawn in or out the detergent supply part and a cartridge identification member storing information about the detergent accommodated in the detergent cartridge. The washing machine may include a cartridge identification device positioned to, with the detergent cartridge drawn into the detergent supply part, recognize the cartridge identification member.

[0013] A washing machine according to an embodiment of the disclosure may include a cabinet. The washing machine may include a tub in the cabinet. The washing machine may include a detergent supply part for supplying a detergent to the tub. The washing machine may include a detergent cartridge accommodating the detergent and configured to be drawn in or out the detergent supply part. The detergent cartridge includes a main body including a first space configured to accommodate air and a second space configured to accommodate a detergent, a partition plate horizontally movable inside the main body and partitioning the first space and the second space, a pair of electric lines extending from the first space to the second space along an inner surface of the main body and spaced apart from each other, and a connector on a side of the partition plate facing the first space and electrically connecting the pair of electric lines. With the detergent cartridge drawn into the detergent supply part, the detergent accommodated in the second space may be supplyable to the tub.

[0014] A method according to an embodiment of the disclosure may be provided for controlling a washing machine that includes (a) a tub, (b) a detergent supply part for supplying a detergent to the tub, (c) a detergent cartridge accommodating the detergent therein, configured to be drawn in or out the detergent supply part, and including a cartridge identification member storing information about the detergent accommodated in the detergent cartridge, and (d) a cartridge identification device positioned to, with the detergent cartridge drawn into the detergent supply part, recognize the cartridge identification member. The method may include (i) with the detergent cartridge drawn into the detergent supply part, recognizing the cartridge identification member through the cartridge identification device; (ii) obtaining the information about the detergent accommodated in the detergent cartridge stored in the cartridge identification member from the recognized cartridge identification member; and (iii) changing a use setting of the

detergent cartridge based on the obtained information about the detergent accommodated in the detergent cartridge.

[0015] According to various embodiments of the disclosure, the washing machine may obtain information (e.g., information about the detergent accommodated in the detergent cartridge) about the detergent cartridge included in a cartridge identification member (e.g., an NFC tag) as a cartridge identification device (e.g., an NFC reader) provided in a detergent supply part recognizes the cartridge identification member provided in the detergent cartridge if the cartridge is drawn in the detergent supply part. The washing machine may automatically change the use settings of the detergent cartridge such as the type of detergent or the amount of detergent put in, based on the obtained information about the detergent cartridge, thereby enhancing user convenience.

[0016] According to various embodiments of the disclosure, the washing machine may measure the resistances of electric lines that vary according to the displacement of a connector moving inside the detergent cartridge as the detergent is discharged, using a current sensor and estimate the remaining amount of detergent accommodated in the detergent cartridge based on the measured resistances, thereby allowing for real-time monitoring of the remaining amount of detergent.

[0017] Effects of the present invention are not limited to the foregoing, and other unmentioned effects would be apparent to one of ordinary skill in the art from the following description. In other words, unintended effects in practicing embodiments of the disclosure may also be derived by one of ordinary skill in the art from example embodiments of the disclosure.

BRIEF DESCRIPTION OF THE DRAWINGS

[0018] The above and other aspects, features, and advantages of certain embodiments of the present disclosure will be more apparent from the following description taken in conjunction with the accompanying drawings, in which:

[0019] FIG. 1 is a perspective view illustrating a washing machine according to an embodiment of the disclosure;

[0020] FIG. 2 is a cross-sectional view illustrating a washing machine according to an embodiment of the disclosure;

[0021] FIG. 3 is a perspective view illustrating an outer appearance of a detergent cartridge according to an embodiment of the disclosure;

[0022] FIG. 4 is a vertical cross-sectional view illustrating a detergent cartridge according to an embodiment of the disclosure;

[0023] FIG. 5 is a vertical cross-sectional view illustrating a detergent cartridge according to an embodiment of the disclosure;

[0024] FIG. 6A is a cross-sectional view schematically illustrating a state in which a detergent cartridge is drawn in a detergent supply part before a detergent is discharged outside according to an embodiment of the disclosure;

[0025] FIG. 6B is a view schematically illustrating a state in which a detergent cartridge is drawn in a detergent supply part when a detergent is discharged outside according to an embodiment of the disclosure;

[0026] FIG. 7 is a block diagram illustrating components connected to a controller of a washing machine according to an embodiment of the disclosure;

[0027] FIG. 8 is a flowchart illustrating a method for controlling a washing machine according to an embodiment of the disclosure;

[0028] FIG. 9 is a view illustrating an automatic use setting of a select cartridge according to the content accommodated in the select cartridge according to an embodiment of the disclosure;

[0029] FIG. 10 is a view illustrating a screen of an external electronic device according to an embodiment of the disclosure;

[0030] FIG. 11 is a flowchart illustrating a method for controlling a washing machine according to an embodiment of the disclosure;

[0031] FIG. 12 is a view exemplarily illustrating an alarm operation of a washing machine per remaining amount of detergent accommodated in a detergent cartridge according to an embodiment of the disclosure;

[0032] FIG. 13 is a perspective view illustrating a washing machine according to an embodiment of the disclosure;

[0033] FIG. 14 is a view illustrating a drawn-out state of a detergent supply device according to an embodiment of the disclosure; and

[0034] FIG. 15 is an exploded perspective view illustrating a detergent supply device according to an embodiment of the disclosure.

DETAILED DESCRIPTION

[0035] It should be appreciated that various embodiments of the present disclosure and the terms used therein are not intended to limit the technological features set forth herein to particular embodiments and include various changes, equivalents, or replacements for a corresponding embodiment.

[0036] With regard to the description of the drawings, similar reference numerals may be used to refer to similar or related elements.

[0037] It is to be understood that a singular form of a noun corresponding to an item may include one or more of the things, unless the relevant context clearly indicates otherwise.

[0038] As used herein, each of such phrases as “A or B,” “at least one of A and B,” “at least one of A or B,” “A, B, or C,” “at least one of A, B, and C,” and “at least one of A, B, or C,” may include all possible combinations of the items enumerated together in a corresponding one of the phrases.

[0039] As used herein, such terms as “1st” and “2nd,” or “first” and “second” may be used to simply distinguish a corresponding component from another, and does not limit the components in other aspect (e.g., importance or order).

[0040] It is to be understood that if an element (e.g., a first element) is referred to, with or without the term “operatively” or “communicatively,” as “coupled with,” “coupled to,” “connected with,” or “connected to” another element (e.g., a second element), it means that the element may be coupled with the other element directly (e.g., wiredly), wirelessly, or via a third element.

[0041] It will be further understood that the terms “comprise” and/or “have,” as used herein, specify the presence of stated features, integers, steps, operations, elements, and/or components, but do not preclude the presence or addition of one or more other features, integers, steps, operations, elements, components, and/or groups thereof.

[0042] It will be understood that when a component is referred to as “connected to,” “coupled to,” “supported on,”

or “contacting” another component, the components may be connected to, coupled to, supported on, or contact each other directly or via a third component.

[0043] Throughout the specification, when one component is positioned “on” another component, the first component may be positioned directly on the second component, or other component(s) may be positioned between the first and second component.

[0044] The term “and/or” may denote a combination(s) of a plurality of related components as listed or any of the components.

[0045] Hereinafter, the working principle and embodiments of the present invention are described with reference to the accompanying drawings.

[0046] A home appliance according to various embodiments may include a clothing treatment device. The washing machine, which is a clothing treatment device, may perform washing, rinsing, spinning, and drying cycles. The washing machine is an example of the clothing treatment device, the clothing treatment device is a concept that encompasses devices for washing clothes (objects to be washed or objects to be dried), devices for drying clothes, and devices capable of performing both washing and drying of clothes.

[0047] The washing machine according to various embodiments may include a top-loading washing machine in which a laundry inlet for inserting or withdrawing laundry faces upward, or a front-loading washing machine in which a laundry inlet faces forward. The washing machine according to various embodiments may include a washing machine having various loading schemes other than the top-loading and front-loading washing machines described above.

[0048] The top-loading washing machine may wash laundry using a water flow generated by a rotating body such as a pulsator. The front-loading washing machine may wash laundry by repeatedly raising and dropping the laundry by rotating the drum. The front-loading washing machine may include a dryer cum washing machine capable of drying laundry accommodated in the drum. The dryer cum washing machine may include a hot air supply device for supplying high-temperature air into the drum and a condensation device for removing moisture from the air discharged from the drum. For example, the dryer cum washing machine may include a heat pump device. The washing machine according to various embodiments may include a washing machine of a washing method other than the above-described washing method.

[0049] The washing machine according to various embodiments may include a housing accommodating various components therein. The housing may be provided in the form of a box having a laundry inlet formed in one side thereof.

[0050] The washing machine may include a door for opening and closing the laundry inlet. The door may be rotatably mounted on the housing by a hinge. At least a portion of the door may be provided to be transparent or translucent so that the inside of the housing is visible.

[0051] The washing machine may include a tub provided inside the housing to store water. The tub may be provided in a substantially cylindrical shape with a tub opening formed in one side thereof, and may be disposed inside the housing so that the tub opening is disposed to correspond to the laundry inlet.

[0052] The tub may be connected to the housing by a damper. The damper may absorb vibrations generated when the drum rotates and attenuate vibrations transferred to the housing.

[0053] The washing machine may include a drum provided to accommodate laundry.

[0054] The drum may be disposed inside the tub so that a drum opening provided in one side corresponds to the laundry inlet and the tub opening. The laundry may be accommodated in the drum or withdrawn from the drum by sequentially passing through the laundry inlet, the tub opening, and the drum opening.

[0055] The drum may perform each operation according to washing, rinsing, and/or spinning cycles while rotating inside the tub. A plurality of through holes may be formed in the cylindrical wall of the drum to allow water stored in the tub to flow into or out of the drum.

[0056] The washing machine may include a driving device configured to rotate the drum. The driving device may include a driving motor and a rotation shaft for transferring the driving force generated by the driving motor to the drum. The rotating shaft may be connected to the drum through the tub.

[0057] The driving device may perform each operation according to washing, rinsing, and/or spinning, or drying cycle by rotating the drum forward or backward.

[0058] The washing machine may include a water supply device configured to supply water to the tub. The water supply device may include a water supply pipe and a water supply valve provided in the water supply pipe. The water supply pipe may be connected to an external water supply source. The water supply pipe may extend from an external water supply source to the detergent supply device and/or the tub. Water may be supplied to the tub via the detergent supply device. Water may be supplied to the tub without passing through the detergent supply device.

[0059] The water supply valve may open or close the water supply pipe in response to an electrical signal from the controller. The water supply valve may allow or block water from being supplied from an external water supply source to the tub. The water supply valve may include, e.g., a solenoid valve that is opened and closed in response to an electrical signal.

[0060] The washing machine may include a detergent supply device configured to supply detergent to the tub. The detergent supply device may include a manual detergent supply device in which the user needs to put a detergent to be used for each washing, and an automatic detergent supply device in which a large amount of detergent is stored and a predetermined amount of detergent is automatically put into the washing. The detergent supply device may include a detergent container for storing the detergent. The detergent supply device may be configured to supply detergent into the tub during a water supply process. Water supplied through the water supply pipe may be mixed with the detergent via the detergent supply device. Water mixed with the detergent may be supplied into the tub. Detergent is used as a term encompassing a detergent for pre-washing, a detergent for main washing, a fabric softener, a bleach, or the like, and the detergent container may be divided into a pre-washing detergent storage area, a main washing detergent storage area, a fabric softener storage area, and a bleach storage area.

[0061] The washing machine may include a drain device configured to discharge water contained in the tub to the outside. The drain device may include a drain pipe extending from a lower portion of the tub to the outside of the housing, a drain valve provided in the drain pipe to open and close the drain pipe, and a pump provided on the drain pipe. The pump may pump water from the drain pipe to the outside of the housing.

[0062] The washing machine may include a control panel disposed on one side surface of the housing. The control panel may provide a user interface for the user and the washing machine to interact with each other. The user interface may include at least one input interface and at least one output interface.

[0063] At least one input interface may convert sensory information received from the user into an electrical signal.

[0064] The at least one input interface may include a power button, an operation button, a course selection dial (or a course selection button), and a washing/rinsing/spinning setting button. The at least one input interface may include, e.g., a tact switch, a push switch, a slide switch, a toggle switch, a micro switch, a touch switch, a touch pad, a touch screen, a jog dial, and/or a microphone.

[0065] The at least one output interface may visually or audibly transfer information related to the operation of the washing machine to the user.

[0066] For example, the at least one output interface may transfer information related to the washing course, the operation time of the washing machine, and the washing setting/rinsing setting/spinning setting to the user. The information about the operation of the washing machine may be output as a screen, an indicator, a voice, etc. The at least one output interface may include, e.g., a liquid crystal display (LCD) panel, a light emitting diode (LED) panel, a speaker, or the like.

[0067] The washing machine may include a communication module for wired and/or wireless communication with an external device.

[0068] The communication module may include at least one of a short-range communication module or a long-range communication module.

[0069] The communication module may transmit data to an external device (e.g., a server, a user device, and/or a home appliance) or may receive data from the external device. For example, the communication module may establish communication with a server and/or the user device and/or a home appliance, and transmit and receive various data.

[0070] To that end, the communication module may support establishing a direct (e.g., wired) communication channel or a wireless communication channel between external devices and performing communication through the established communication channel. According to an embodiment, the communication module may include a wireless communication module (e.g., a cellular communication module, a short-range wireless communication module, or a global navigation satellite system (GNSS) communication module) or a wired communication module (e.g., a local area network (LAN) communication module or a power line communication (PLC) module). A corresponding one of these communication modules may communicate with the external device via a first network (e.g., a short-range communication network, such as Bluetooth™, wireless-fidelity (Wi-Fi) direct, or infrared data association (IrDA))

or a second network (e.g., a long-range communication network, such as a legacy cellular network, a 5G network, a next-generation communication network, the Internet, or a computer network (e.g., local area network (LAN) or wide area network (WAN))). These various types of communication modules may be implemented as a single component (e.g., a single chip), or may be implemented as multi components (e.g., multi chips) separate from each other.

[0071] The short-range wireless communication module may include, but is not limited to, a Bluetooth communication module, a BLE communication module, a near field communication module, a WLAN (Wi-Fi) communication module, a Zigbee communication module, an infrared data association (IrDA) communication module, a Wi-Fi direct (WFD) communication module, an ultrawideband (UWB) communication module, an Ant+ communication module, and a microwave (uWave) communication module.

[0072] The long-range communication module may include a communication module that performs various types of long-range communication, and may include a mobile communication unit. The mobile communication unit transmits and receives a wireless signal to and from at least one of a base station, an external terminal, and a server over a mobile communication network.

[0073] In an embodiment, the communication module may communicate with an external device such as a server, a user equipment, another home appliance, or the like through an access point (AP). The AP may connect the local area network (LAN) to which the washing machine or the user equipment is connected to the wide area network (WAN) to which the server is connected. The washing machine or the user device may be connected to the server through the wide area network (WAN). The controller may control various components (e.g., a driving motor and a water supply valve) of the washing machine. The controller may control various components of the washing machine to perform at least one cycle including water supply, washing, rinsing, and/or spinning according to the user input. For example, the controller may control the driving motor to adjust the rotational speed of the drum, or control the water supply valve of the water supply device to supply water to the tub.

[0074] The controller may include hardware such as a CPU or a memory, and software such as a control program. For example, the controller may include an algorithm for controlling operations of the components in the washing machine, at least one memory for storing data in the form of a program, and at least one processor for performing the above-described operations using the data stored in the at least one memory. The memory and the processor may be implemented as separate chips, respectively. The processor may include one or more processor chips or may include one or more processing cores. The memory may include one or more memory chips or may include one or more memory blocks. Further, the memory and the processor may be implemented as a single chip.

[0075] FIG. 1 is a perspective view illustrating a washing machine according to an embodiment of the disclosure.

[0076] FIG. 2 is a cross-sectional view illustrating a washing machine according to an embodiment of the disclosure.

[0077] Specifically, FIG. 1 is a perspective view illustrating a state in which a door 40 of a washing machine 1 according to an embodiment is opened.

[0078] Referring to FIGS. 1 and 2, a washing machine 1 according to an embodiment may include a cabinet 10 forming an outer appearance and accommodating various components therein, a tub 20 provided inside the cabinet 10 to store mixed water in which water and detergent are mixed, a drum 30 provided inside the tub 20 to accommodate laundry and rotate, a door 40 rotatably coupled to the cabinet 10, a driving device 60 rotating the drum 30, a detergent supply part 90 for supplying the detergent to the tub 20, and a detergent cartridge 100 provided to be drawn in/out the detergent supply part 90.

[0079] According to an embodiment, the cabinet 10 may have a hexahedral shape as a whole. In an embodiment, the cabinet 10 may be formed in a box shape with an open front.

[0080] According to an embodiment, the cabinet 10 may include a front panel 11a provided in the open front (e.g., the +x-axis direction), side panels 11b and 11c disposed on two opposite side surfaces (e.g., the surfaces facing in the +y-axis directions) rearward of the front panel 11a, a rear panel 11d provided at the rear (e.g., the -x-axis direction) of the cabinet 10, and a top cover 11e provided on an upper surface (e.g., the surface facing in the +z-axis direction) thereof. An example in which in the cabinet 10, the rear panel 11d and two opposite side panels 11b and 11c are integrally formed is illustrated, but the disclosure is not limited thereto.

[0081] According to an embodiment, an opening 12 may be formed in the front panel 11a to allow laundry to be put into the drum 30. Openings may be formed in the tub 20 and the drum 30 to allow laundry to be put in or out of the cabinet 10, and the openings of the tub 20 and the drum 30 may be positioned to correspond to the opening 12 of the front panel 11a.

[0082] According to an embodiment, the cabinet 10 may include a base 13 provided at the bottom. In an embodiment, the base 13 may include a lower frame 13a formed to cover a lower portion of the cabinet 10 and a lower reinforcing frame 13b formed to extend from the lower frame 13a.

[0083] According to an embodiment, a water supply device 14 including a water supply pipe 14a for supplying washing water to the detergent supply part 90 and/or the tub 20, and a water supply valve 14b for controlling water supply of the water supply pipe 14a may be provided above the tub 20. Further, the water supply device 14 may further include a water supply nozzle (e.g., the water supply nozzle 14c of FIGS. 6A and 6B) for jetting washing water into the detergent supply part 90. The water supply device 14 may be referred to as a water supply unit.

[0084] According to an embodiment, the detergent supply part 90 having an opening (or inlet) 90a through which the detergent cartridge 100 is drawn in/out may be provided above the tub 20. In an embodiment, a cover 15 may be provided in the front of the washing machine 1 to open/close the opening 90a. The cover 15 may be rotatably connected to the front side (e.g., the front panel 11a) of the washing machine 1.

[0085] According to an embodiment, the detergent supply part 90 may be connected to the water supply pipe 14a at the rear. In an embodiment, the detergent supply part 90 may be connected to an upper portion of the tub 20 through a supply pipe 16. The water supplied through the water supply pipe 14a may be mixed with the detergent via the detergent supply device 90 through the water supply nozzle 14c, and

the mixed water in which the water and the detergent are mixed may be supplied into the tub 20 through the supply pipe 16.

[0086] According to an embodiment, a diaphragm 17 connecting the opening 12 of the front panel 11a and the opening of the tub 20 may be provided between the front panel 11a and the tub 20.

[0087] According to an embodiment, the tub 20 may store mixed water in which water and a detergent are mixed, and may be formed in a cylindrical shape as a whole. In an embodiment, the tub 20 may be fixed inside the cabinet 10. Although not specifically shown in the drawings, a drain device (not shown) including a drain pipe (not shown) for draining water inside the tub 20 and a drain valve (not shown) may be installed under the tub 20. The drain device may be referred to as a drain part.

[0088] According to an embodiment, the tub 20 may be provided to be elastically supported from the cabinet 10 by an upper spring (not shown) and lower dampers 80. When vibration generated when the drum 30 rotates is transferred to the tub 20 and the cabinet 10, the spring and dampers 80 may absorb vibration energy between the tub 20 and the cabinet 10 to attenuate the vibration.

[0089] According to an embodiment, the drum 30 may perform a washing cycle of raising and dropping laundry while rotating inside the tub 20. A lifter 31 and/or a plurality of through-holes 32 may be provided in the inner circumferential surface of the drum 30. The lifter 31 may lift the laundry while the drum 30 rotates so that the laundry repeatedly rises and falls, thereby evenly washing laundry on several surfaces thereof. The through hole 32 may be a passage formed so that the water or washing water accommodated in the tub 20 flows into the drum 30, or the water or washing water inside the drum 30 is discharged to the outside.

[0090] According to an embodiment, the door 40 may be rotatably coupled to the front panel 11a of the cabinet 10. In an embodiment, the door 40 may rotate to open and close the opening 12 of the front panel 11a. In an embodiment, the door 40 may include a door frame 41 and a door glass 43. The door glass 43 may be formed of a transparent tempered glass material to see through the inside of the cabinet 10.

[0091] According to an embodiment, a control panel 50 including an input device 51 for receiving an operation command of the washing machine 1 from the user and an output device 52 for outputting operation information about the washing machine 1 may be disposed on the front side of the cabinet 10. The input device 51 may be implemented as an input means such as a dial switch as shown in the drawings, but the disclosure is not limited thereto. The output device 52 may be implemented as an output means in the form of a display as illustrated in the drawings, but the disclosure is not limited thereto.

[0092] According to an embodiment, the driving device 60 may be disposed behind the drum 30. In an embodiment, the driving device 60 may be provided to rotate the drum 30 by transferring the driving force generated by the motor 61 to the drum 30 through the rotation shaft 67. In an embodiment, the motor 61 may include a fixed stator 63 and a rotor 65 that rotates by electromagnetically interacting with the stator 63 to convert an electric force into a rotational force.

[0093] According to an embodiment, a spray nozzle 70 connected to the circulation hose 71 to spray the circulating mixed water into the drum 30 may be provided on one side

of the diaphragm 17. In an embodiment, the spray nozzle 70 may be configured to evenly spray mixed water to the entire laundry accommodated in the drum 30. In an embodiment, the circulation hose 71 may be connected to a circulation pump 73 for pumping mixed water stored under the tub 20.

[0094] According to an embodiment, the detergent cartridge 100 may be disposed on the front upper portion of the tub 20. In an embodiment, the detergent cartridge 100 may be drawn into the washing machine 1 or withdrawn out of the washing machine 1 through the opening 90a of the detergent supply part 90. In an embodiment, the detergent cartridge 100 may store a detergent for washing therein. The detergent may include a general detergent, a neutral detergent, an infant detergent, or a softener. In an embodiment, the detergent cartridge 100 may automatically supply detergent to the inside (e.g., the tub 20) of the washing machine 1 by interaction with a nozzle pressurizer (e.g., the nozzle pressurizer 930 of FIGS. 6A and 6B) described below when drawn in the detergent supply part 90. The detergent cartridge 100 may also be referred to as an automatic detergent container.

[0095] Hereinafter, the structure and/or shape of the detergent cartridge 100 is described in detail with reference to FIGS. 3 and 4.

[0096] FIG. 3 is a perspective view illustrating an outer appearance of a detergent cartridge according to an embodiment of the disclosure.

[0097] FIG. 4 is a vertical cross-sectional view illustrating a detergent cartridge according to an embodiment of the disclosure.

[0098] Referring to FIGS. 3 and 4, the detergent cartridge 100 may include a pair of left and right detergent cartridges 100a and 100b drawn in the detergent supply part (e.g., the detergent supply part 90 of FIG. 2).

[0099] According to an embodiment, the pair of detergent cartridges 100a and 100b may include a first cartridge 100a in which a detergent is accommodated and a second cartridge 100b in which a detergent or a softener is accommodated. The first cartridge 100a may be referred to as a normal cartridge, and the second cartridge 100b may be referred to as a select cartridge.

[0100] According to an embodiment, the pair of detergent cartridges 100a and 100b may selectively contact a nozzle pressurizer (e.g., the nozzle pressurizer 930 of FIGS. 6A and 6B) to be described below, and may accommodate a detergent for automatic input therein.

[0101] According to an embodiment, the pair of detergent cartridges 100 may include a main body 110 in which detergent is accommodated, a partition plate 120 for partitioning the inside of the main body 110 into a first space 110a and a second space 110b, a nozzle device 130 for discharging detergent to the outside, a pair of electric lines 140 spaced apart from each other, a connector 150 for electrically connecting the pair of electric lines 140, and a cartridge identification member 160 including information about the detergent accommodated in the detergent cartridge 100.

[0102] According to an embodiment, the main body 110 of the detergent cartridge 100 may have a rectangular parallelepiped shape as a whole. According to an embodiment, the main body 110 may include a first space 110a filled with air and a second space 110b filled with detergent. The first space 110a and the second space 110b may be divided from each other by the partition plate 120. The first space 110a

and the second space 110b may be arranged in a horizontal direction, and the first space 110a may be positioned in front of the second space 110b. The first space 110a may be referred to as an air area, and the second space 110b may be referred to as a detergent area. The air filling the first space 110a may continuously pressurize the partition plate 120, and when the nozzle device 130 and the nozzle pressurizer 930, which are described below, interact, the detergent filling the second space 110b may be discharged to the outside.

[0103] According to an embodiment, the partition plate 120 may be horizontally movable inside the main body 110. In an embodiment, the partition plate 120 may partition the first space 110a and the second space 110b. In an embodiment, the partition plate 120 may move in the horizontal direction according to the remaining amount of detergent accommodated in the second space 110b. For example, the partition plate 120 may move horizontally to the right as the detergent accommodated in the second space 110b decreases.

[0104] According to an embodiment, the nozzle device 130 may be provided to discharge the detergent accommodated in the second space 110b of the main body 110 to the outside. In an embodiment, the nozzle device 130 may include a nozzle body 131, an elastic member 132 providing an elastic force to the nozzle body 131, and a fixing portion 133 fixing the elastic member 132 to the second area 110b.

[0105] According to an embodiment, the nozzle body 131 may include a flow path 1311 through which the detergent flows and a pair of stoppers 1312 vertically extending from a side surface of the nozzle body 131.

[0106] In an embodiment, the nozzle body 131 may penetrate the rear wall 113 of the main body 110 of the detergent cartridge 100. An inlet end of the flow path 1311 of the nozzle body 131 may be open in the upper side of the nozzle body 131, and an outlet end of the flow path 1311 of the nozzle body 131 may be bent and extended from the inlet end to allow the outlet end to be open in the side surface of the nozzle body 131. The nozzle body 131 may selectively communicate with the second space 110b according to the degree of external force applied by the nozzle pressurizer 930, and may discharge the detergent to the outside through the flow path 1311. For example, when no external force is applied to the nozzle device 130, the inlet end of the nozzle body 131 may be surrounded by the rear wall 113 of the main body 110 of the detergent cartridge 100 not to communicate with the second space 110b. For example, when an external force is applied to the nozzle device 130, the inlet end of the nozzle device 1311 may communicate with the second space 110b according to the degree of compression of the elastic member 132 by the nozzle pressurizer 930, and the detergent accommodated in the second space 110b may be discharged to the outside through the flow path 1311.

[0107] According to an embodiment, the elastic member 132 may be elastically connected to the nozzle body 131. The elastic member 132 may continuously provide an elastic force to the nozzle body 131. The elastic member 132 may communicate the flow path 1311 of the nozzle body 131 with the second space 110b while changing the amount of compression according to the degree of application of the external force to the nozzle device 130.

[0108] According to an embodiment, the fixing portion 133 may accommodate the elastic member 132 therein and may be fixed to the rear wall 113 of the main body 110 of

the detergent cartridge **100**. In an embodiment, an opening **133a** may be formed in one side (e.g., the front side) of the fixing portion **133**, and the detergent accommodated in the second space **110b** may be introduced into the fixing portion **133** through the opening **133a**.

[0109] According to an embodiment, the pair of electric lines **140** may be provided on the inner surface of the main body **110** of the detergent cartridge **100**. In an embodiment, the pair of electric lines **140** may include a first electric line **140a** and a second electric line **140b** spaced apart from each other on the inner surface of the main body **110**. The pair of electric lines **140** may be collectively referred to as a resistance line or a lead line. In an embodiment, the first electric line **140a** may have one end **143a** provided on the lower surface **112** of the main body **110** in the form of an electrode, and may include a 1-1th extension **141a** extending in a vertical direction along the one end **143a** and a 1-2th extension **142a** extending in a horizontal direction from the 1-1th extension **141a**. In an embodiment, the second electric line **140b** may have one end **143b** provided on the lower surface **112** of the main body **110** in the form of an electrode, and may include a 2-1th extension **141b** extending in a vertical direction along the one end **143b** and a 2-2th extension **142b** extending in a horizontal direction from the 2-1th extension **141b**.

[0110] According to an embodiment, the connector **150** may electrically connect the first electric line **140a** and the second electric line **140b**. In an embodiment, the connector **150** may be fixed to one side (e.g., the front side) of the partition plate **122** and may move together with the partition plate **120**. When the connector **150** moves to the right, the total length of the first electric line **140a** and the second electric line **140b** connected by the connector **150** may increase, and thus the resistance of the circuit formed by the first electric line **140a** and the second electric line **140b** may also increase. In an embodiment, the connector **150** may include a connector body **151** fixed to one side of the partition plate **120**, and connector members **1521** and **1522** fixed to the connector body **151** and elastically contacting the first electric line **140a** and the second electric line **140b**, respectively. The connector members **1521** and **1522** may be provided, e.g., in a C-clip shape having an elastically deformable conductive material, but the disclosure is not limited thereto.

[0111] According to an embodiment, the cartridge identification member **160** may store information about the detergent accommodated in the detergent cartridge **100**. The information about the detergent may include, e.g., information about the type of detergent, the standard amount of detergent, the manufacturer of detergent, and the concentration level of detergent. Here, the standard amount of detergent may mean the input amount of detergent corresponding to the amount (load) of laundry. The cartridge identification member **160** may be implemented, e.g., in the form of an NFC tag, but the disclosure is not limited thereto. Hereinafter, for convenience of description, the cartridge identification member **160** will be referred to as an NFC tag. In an embodiment, the NFC tag **160** may be disposed on the upper side **111** of the main body **110** of the detergent cartridge **100**. For example, the NFC tag **160** may be disposed on each of the upper sides of the first cartridge **100a** and the second cartridge **100b**. In an embodiment, the NFC tag **160** may be positioned adjacent to an NFC reader (the detergent container identification device (hereinafter, referred to as the

NFC reader **920**) of FIGS. **6A** and **6B**) to be described below. The NFC tag **160** may support the detergent use setting of the detergent cartridge **100** to be automatically set by transmitting the information about the detergent stored by the NFC reader **920** to the washing machine **1**.

[0112] FIG. **5** is a vertical cross-sectional view illustrating a detergent cartridge according to an embodiment of the disclosure.

[0113] Referring to FIG. **5**, a pair of detergent cartridges **100'** according to an embodiment may include a main body **110** in which detergent is accommodated, a partition plate **120** for partitioning the inside of the main body **110** into a first space **110a** and a second space **110b**, a nozzle device **130** for discharging the detergent to the outside, a pair of electric lines **140** spaced apart from each other, a connector **150** for electrically connecting the pair of electric lines **140'**, and an NFC tag **160** including information about the detergent accommodated in the detergent cartridge **100**.

[0114] Below, a description of portions overlapping those described with reference to FIGS. **3** and **4** is omitted.

[0115] According to an embodiment, the pair of electric lines **140a'** and **140b'** may be provided on the inner surface of the main body **110** of the detergent cartridge **100'**. In an embodiment, the pair of electric lines **140a'** and **140b'** may include a first electric line **140a'** and a second electric line **140b'** spaced apart from each other on the inner surface of the main body **110**. The pair of electric lines **140a'** and **140b'** may be collectively referred to as a resistance line or a lead line. In an embodiment, the pair of electric lines **140a'** and **140b'** may have various pattern shapes. The pair of electric lines **140a'** and **140b'** may have, e.g., an irregular pattern (see FIG. **5**) or a wavy pattern. Compared with the pair of electric lines **140a** and **140b** illustrated in FIG. **4**, the pair of electric lines **140a'** and **140b'** illustrated in FIG. **5** have various patterns, so that the total length of the electric lines may increase compared to the horizontal length of the electric line, and thus a change in resistance according to the movement of the connector **150** may be measured more precisely.

[0116] In an embodiment, the first electric line **140a'** may have one end **143a** provided on the lower side **112** of the main body **110** in the form of an electrode, and may include a 1-1th extension **141a** extending in a vertical direction along the one end **143a** and a 1-2th extension **144a** extending in a horizontal direction from the 1-1th extension **141a**. In an embodiment, the second electric line **140b'** may have one end **143b** provided on the lower surface **112** of the main body **110** in the form of an electrode, and may include a 2-1th extension **141b** extending in a vertical direction along the one end **143b** and a 2-2th extension **144b** extending in a horizontal direction from the 2-1th extension **141b**.

[0117] FIG. **6A** is a cross-sectional view schematically illustrating a state in which a detergent cartridge is drawn in a detergent supply part before a detergent is discharged outside according to an embodiment of the disclosure.

[0118] FIG. **6B** is a view schematically illustrating a state in which a detergent cartridge is drawn in a detergent supply part when a detergent is discharged outside according to an embodiment of the disclosure.

[0119] FIGS. **6A** and **6B** are views illustrating an operation of detecting whether the detergent cartridge **100** is mounted on the detergent supply part **90** or an operation of detecting the remaining amount of detergent by measuring the resistance of electric lines **140a** and **140b** that change

according to the movement of the connector 150 when the detergent accommodated in the detergent cartridge 100 is discharged. Specifically, FIG. 6A is a view illustrating a state before detergent is discharged, and FIG. 6B is a view illustrating a state after detergent is discharged.

[0120] Referring to FIGS. 6A and 6B, a pair of detergent cartridges 100 according to an embodiment may include a main body 110 in which detergent is accommodated, a partition plate 120 for partitioning the inside of the main body 110 into a first space 110a and a second space 110b, a nozzle device 130 for discharging the detergent to the outside, a pair of electric lines 140a and 140b spaced apart from each other, a connector 150 for electrically connecting the pair of electric lines 140a and 140b, and a cartridge identification member (hereinafter, an NFC tag) 160 including information about the detergent accommodated in the detergent cartridge 100.

[0121] According to an embodiment, the detergent supply part 90 may include an opening 90a for inputting the detergent cartridge 100 into the washing machine and a detergent supply part housing 910 defining a space 90b in which the detergent cartridge 100 is accommodated.

[0122] According to an embodiment, the washing machine (e.g., the washing machine 1 of FIGS. 1 and 2) may include a cartridge identification device 920 for obtaining information about the detergent stored in the NFC tag 160. The cartridge identification device 920 may be implemented in the form of an NFC reader for recognizing an NFC tag. Hereinafter, for convenience of description, the cartridge identification device 920 is referred to as an NFC reader 920. In an embodiment, the NFC reader 920 may be positioned adjacent to the NFC tag 920. For example, the NFC reader 920 may be disposed on the upper inner surface 911 of the detergent supply part 90 so that at least a portion thereof faces the NFC tag 160 when the detergent cartridge 100 is drawn in the detergent supply part 90. The NFC reader 920 may be electrically connected to a controller (e.g., the controller 190 of FIG. 7) of the washing machine 1.

[0123] According to an embodiment, the washing machine 1 may include a current sensor 183 (e.g., the second current sensor 183 of FIG. 7) for measuring resistances of the pair of electric lines 140a and 140b.

[0124] In an embodiment, the current sensor 183 may be electrically connected to each of the pair of electric lines 140a and 140b when the detergent cartridge 100 is drawn in the detergent supply 90, and may constitute a closed circuit together with the pair of electric lines 140a and 140b and the connector 150. In an embodiment, the current sensor 183 may include a pair of electrodes 183a and 183b that contact the respective end portions 143a and 143b of the pair of electric lines 140a and 140b when the detergent cartridge 100 is drawn in the detergent supply part 90. In an embodiment, the pair of electrodes 183a and 183b may be provided at positions facing the respective end portions 143a and 143b of the pair of electric lines 140a and 140b on the bottom surface 912 of the detergent supply part 90 when the detergent cartridge 100 is drawn in the detergent supply part 90.

[0125] When the detergent cartridge 100 is drawn in the detergent supply part 90, the respective end portions 143a and 143b of the electric lines 140a and 140b provided in the detergent cartridge 100 and the electrodes 183a and 183b of the current sensor 183 come into contact with each other, so that current flows along the electric lines 140a and 140b and

resistances of the electric lines 140a and 140b may be measured. In this case, the resistance (or current) measured by the current sensor 183 may be compared with a preset setting value to determine whether the detergent cartridge 100 is mounted in the detergent supply part 90.

[0126] According to an embodiment, the washing machine 1 may include a nozzle pressurizer 930 for discharging the detergent contained in the detergent cartridge 100. In an embodiment, the nozzle pressurizer 930 may be positioned behind the detergent cartridge 100 when the detergent cartridge 100 is drawn in the detergent supply part 90. For example, the nozzle pressurizer 930 may be disposed to face the nozzle device 130 of the detergent cartridge 100 when the detergent cartridge 100 is drawn in the detergent supply part 90.

[0127] In an embodiment, by pressurizing the nozzle device 130, the nozzle pressurizer 930 may communicate the flow path 1311 of the nozzle device 130 with the second space 110b in which the detergent is accommodated to discharge the detergent to the outside.

[0128] In an embodiment, the nozzle pressurizer 930 may include a pair of pressing portions 931 and 932 disposed to face the pair of stoppers 1312 of the nozzle device 130. In an embodiment, the nozzle pressurizer 930 may be configured to be horizontally movable. In an embodiment, when the nozzle pressurizer 930 moves horizontally toward the nozzle device 130, the pair of pressing portions 932 and 932 may contact and pressurize the stopper 1312 of the nozzle device 130 to move the nozzle device 130 into the detergent cartridge 100. In this case, the flow path 1311 of the nozzle device 130 may communicate with the second space 110b in which the detergent is accommodated, so that the detergent may be discharged to the outside.

[0129] When the detergent is discharged out of the detergent cartridge 100 by the operation of the nozzle pressurizer 930, the volume of the second space 110b is reduced by the amount of the discharged detergent, and the partition plate 120 is horizontally moved to the right. When the partition plate 120 is moved, the connector 150 is also moved to the right, and the length of circuit comprising the electric lines 140a and 140b and the connector 150 is increased. In this case, the resistance of the circuit comprising electric lines 140a and 140b and the connector 150 increases, and the resistance measured through the current sensor 183 increases. As such, the washing machine 1 may monitor the remaining amount of detergent present in the detergent cartridge 100 in real time by measuring the resistances of the electric lines 140a and 140b which are changed by the movement of the partition plate 120 or the connector 150 through the current sensor 183 when the detergent is discharged according to the operation of the nozzle pressurizer 930.

[0130] FIG. 7 is a block diagram illustrating components connected to a controller of a washing machine according to an embodiment of the disclosure.

[0131] Referring to FIG. 7, a washing machine 1 according to an embodiment may include an input device 51, a sensor part 180, an NFC tag 160, a communication part 170, a controller 190, a memory 191, a driving device 60, an output device 52, a nozzle pressurizer 930, or a water supply device 14.

[0132] Therefore, a description of portions overlapping those described with reference to FIGS. 1 to 6 is omitted.

[0133] According to an embodiment, the sensor part 180 may include a spin-dry sensor 181, a first current sensor 182, a second current sensor 183, a door sensor 184, or an optical sensor 185, but this is illustrative and the disclosure is not limited thereto.

[0134] According to an embodiment, the spin-dry sensor 181 is a sensor provided to detect the water level in the tub 20, and may be provided to identify the progress of spin-drying when performing the spin-dry cycle. The controller 190 may receive information about the water level in the tub 20 from the spin-dry sensor 181 and reference it for the spin-dry cycle control.

[0135] According to an embodiment, the first current sensor 182 may be provided to detect a current flowing through the motor 61 of the driver 60. The controller 190 may estimate the torque value applied to the motor 61 using the current value measured by the first current sensor 182. The controller 190 may estimate the amount of laundry in the drum (e.g., the drum 30 of FIG. 2) based on, e.g., the estimated torque value.

[0136] According to an embodiment, the second current sensor 183 may be provided to detect resistances (or currents) flowing through the electric lines 140a and 140b of the detergent cartridge 100. The controller 190 may estimate the moving displacement of the partition plate 120 or the connector 150 using the current value measured by the second current sensor 183. For example, the controller 190 may estimate the remaining amount of the detergent accommodated in the detergent cartridge 100 based on the estimated moving displacement.

[0137] According to an embodiment, the door sensor 184 may be provided to determine whether the user closes the door 40 before the controller 190 performs the washing operation.

[0138] According to an embodiment, the optical sensor 185 may be provided to identify the degree of contamination inside the washing machine 1 by measuring the reflectance of a partial area of the inner surface of the washing machine 1 (e.g., the tub 20). The controller 190 (or the processor) may receive the degree of contamination in the washing machine 1 from the optical sensor 145, determine whether washing is performed (e.g., whether it is necessary to wash the tub), and transmit the determination results to the user through the output device 52 or the like, or execute at least one instruction preset (or stored) in the memory 191 to perform washing.

[0139] According to an embodiment, the communication part 170 may include one or more modules that enable wireless communication between the washing machine 1 and a wireless communication system, between the washing machine 1 and another device, or between the washing machine 1 and an external server. According to an embodiment, the communication part 170 may include one or more modules for connecting the washing machine 1 to one or more networks. According to an embodiment, the communication part 170 may include at least one of a mobile communication module, a wireless Internet module, a short-range communication module, or a location information module.

[0140] According to an embodiment, the mobile communication module may transmit/receive a wireless signal to/from at least one of a base station, an external terminal, or a server on a mobile communication network established according to technical standards or communication methods

for mobile communication. The wireless signals may include, e.g., voice call signals, video call signals, or other various types of data according to transmission/reception of text/multimedia messages.

[0141] According to an embodiment, the wireless Internet module may be, but is not limited to, e.g., WLAN (Wireless LAN), Wi-Fi (Wireless-Fidelity), Wi-Fi Direct, DLNA (Digital Living Network Alliance), WiBro (Wireless Broadband), WiMAX (World Interoperability for Microwave Access), HSDPA (High Speed Downlink Packet Access), HSUPA (High Speed Uplink Packet Access), LTE (Long Term Evolution), LTE-A (Long Term Evolution-Advanced), or 5G. Data may be transmitted and received according to at least one wireless Internet technology in the scope encompassing Internet technologies even not enumerated above.

[0142] According to an embodiment, the short-range communication module may be intended for, e.g., short-range communication and may support short-range communication using at least one of Bluetooth, radio frequency identification (RFID), infrared data association (IrDA), ultra-wideband (UWB), ZigBee, near-field communication (NFC), Wi-Fi, Wi-Fi Direct, or wireless universal serial bus (USB) technology. The short-range communication module may support, e.g., wireless communication between the washing machine 1 and a wireless communication system, between the washing machine 1 and another device, or between the washing machine 1 and a network in which the other device is positioned through a short-range wireless communication network. Here, the wireless local area network may be a wireless personal area network.

[0143] According to an embodiment, the location information module may be, e.g., a global positioning system (GPS) module or a Wi-Fi module as a module for obtaining the location of the washing machine 1. When the washing machine 1 utilizes the GPS module, the washing machine 1 may receive information about the location of the washing machine 1 using the signal transmitted from the GPS satellite. When the washing machine 1 utilizes the Wi-Fi module, the washing machine 1 may receive information about the location of the washing machine 1 based on information about a wireless access point (AP) that transmits and receives a wireless signal to and from the Wi-Fi module.

[0144] According to an embodiment, the controller 190 may control the overall operation of the washing machine 1. To that end, the controller 190 may include one or more of a central processing unit (CPU), an application processor (AP), or a communication processor (CP). The controller 190 may be, e.g., a micro control unit (MCU).

[0145] According to an embodiment, the controller 190 may control hardware or

[0146] software components connected to the controller 190 by driving, e.g., an operating system or an application program, and may perform various data processing and operations. Further, the controller 190 may load and process a command or data received from at least one of other components in a volatile memory, and store various data in a non-volatile memory.

[0147] According to an embodiment, the controller 190 may control the washing cycle using, e.g., a signal received from the input device 51 or the communication part 170 (e.g., the NFC reader 920). Specifically, the controller 190 may control at least one of a washing cycle, a rinsing cycle, a spin-dry cycle, or a drying cycle. The controller 190 may control the operation of the driving device 60, the nozzle

pressurizer 930, or the water supply device 14 to control at least one of the washing cycle, the rinsing cycle, the spin-dry cycle, or the drying cycle. At this time, the controller 190 may receive operation information about the washing machine 1 from the sensor part 180 and control feedback.

[0148] According to an embodiment, the memory 191 may store data supporting various functions of the washing machine 1. The memory 191 may store, e.g., a plurality of application programs (or applications) used in the washing machine 1, or data or instructions for operating the washing machine 1. At least some of these applications may be downloaded from an external server through wireless communication. Further, at least some of the application programs may be stored in the memory 191 from the time of shipping to perform default functions of the washing machine 1. For example, the application program may be stored in the memory 191 and driven to perform an operation (or function) of the washing machine 1 by the controller 190. According to an embodiment, the memory 191 may be included as a part of the controller 190.

[0149] FIG. 8 is a flowchart illustrating a method for controlling a washing machine according to an embodiment of the disclosure.

[0150] FIG. 9 is a view illustrating an automatic use setting of a select cartridge according to the content accommodated in the select cartridge according to an embodiment of the disclosure.

[0151] FIG. 10 is a view illustrating a screen of an external electronic device according to an embodiment of the disclosure.

[0152] Referring to FIGS. 8 to 10, the washing machine 1 according to an embodiment may obtain information about the detergent stored in the detergent cartridges 100a and 100b by recognizing the NFC tag 160 provided in the detergent cartridges 100, e.g., the first cartridge (hereinafter, referred to as a normal cartridge) 100a and the second cartridge (hereinafter, referred to as a select cartridge) 100b, and may automatically change the use settings of the detergent cartridges 100 based on the obtained information about the detergent.

[0153] According to an embodiment, the washing machine 1 may determine whether the normal cartridge 100a is mounted on the detergent supply part 90 (805). The washing machine 1 may determine whether the normal cartridge 100a is drawn in the detergent supply part 90 based on the resistance (or the current value) detected by the current sensor (e.g., the second current sensor) 183 provided in the detergent supply part 90. For example, when the normal cartridge 100a is drawn in the detergent supply part 90, the electric lines 140a and 140b provided in the normal cartridge 100a and the current sensor 183 are electrically connected to each other, and thus the current sensor 183 may measure the resistances of the electric lines 140a and 140b. In this case, the washing machine 1 may compare the resistance measured through the current sensor 183 with a preset resistance to determine whether the normal cartridge 100a is mounted on the detergent supply part 90.

[0154] According to an embodiment, when the normal cartridge 100a is mounted on the detergent supply part 90 (YES in 805), the washing machine 1 may recognize the NFC tag 160 provided in the normal cartridge 100a through the NFC reader 920 provided in the detergent supply part 90 (810). For example, when the normal cartridge 100a is drawn in the detergent supply part 90, the NFC reader 920

may recognize the NFC tag 160 as the NFC reader 920 and the NFC tag 160 of the normal detergent container 120a come adjacent to each other, and the washing machine 1 may obtain information about the detergent of the normal cartridge 100a stored in the NFC tag 160.

[0155] According to an embodiment, the washing machine 1 may determine whether the information about the detergent of the normal cartridge 100a is changed from the existing information (815). For example, the washing machine 1 may compare the information about the detergent of the normal cartridge 100a obtained in operation 810 with the information about the detergent of the normal cartridge 100a stored in the memory 191 to determine whether the information about the normal cartridge 100a is changed.

[0156] According to an embodiment, when the information about the detergent of the normal cartridge 100a is changed (YES in 815), the washing machine 1 may change the use settings of the normal cartridge 100a based on the detergent information included in the NFC tag 160 of the mounted normal cartridge 100a (820). For example, the washing machine 1 may detect a change in the detergent accommodated in the normal cartridge 100a through the NFC tag 160 and the NFC reader 920, and automatically change the use settings of the normal cartridge 100a based on the information about the new detergent of the normal cartridge 100a. The use settings of the normal cartridge 100a may include, e.g., information about the detergent such as the type of detergent accommodated in the normal cartridge 100a and the input amount of the detergent.

[0157] According to an embodiment, the washing machine 1 may store the current use settings of the changed normal cartridge 100a in the memory 191 (825). Thereafter, the washing machine 1 may perform operation 830 described below. However, the disclosure is not limited thereto, and the operations illustrated in FIG. 8 may be performed simultaneously.

[0158] According to an embodiment, when the normal cartridge 100a is not mounted on the detergent supply part 90 (NO in 805), the washing machine 1 may determine whether the select cartridge 100b is mounted on the detergent supply part 90 (830). The washing machine 1 may determine whether the select cartridge 100b is drawn in the detergent supply part 90 based on the resistance (or the current value) detected by the current sensor 183 provided in the detergent supply part 90. For example, when the select cartridge 100b is drawn in the detergent supply part 90, the electric lines 140a and 140b provided in the select cartridge 100b are electrically connected to the current sensor 183, and thus the current sensor 183 may measure the resistances of the electric lines 140a and 140b. In this case, the washing machine 1 may compare the resistance measured through the current sensor 183 with a predetermined resistance to determine whether the select cartridge 100b is mounted on the detergent supply part 90.

[0159] According to an embodiment, when the select cartridge 100b is mounted on the detergent supply part 90 (YES in 830), the washing machine 1 may recognize the NFC tag 160 provided in the select cartridge 100b through the NFC reader 920 provided in the detergent supply part 90 (835). For example, when the select cartridge 100b is drawn in the detergent supply part 90, the NFC reader 920 may recognize the NFC tag 160 as the NFC reader 920 and the NFC tag 160 of the select cartridge 100b come adjacent to

each other, and the washing machine 1 may obtain information about the detergent of the select cartridge 100b stored in the NFC tag 160.

[0160] According to an embodiment, the washing machine 1 may determine whether the detergent information about the select cartridge 100b is changed from the existing information (840). For example, the washing machine 1 may compare the information about the detergent of the select cartridge 100b obtained in operation 835 with the information about the detergent of the select cartridge 100b stored in the memory 191 to determine whether the information about the select cartridge 100b is changed.

[0161] According to an embodiment, when the information about the detergent of the select cartridge 100b is changed (YES in 840), the washing machine 1 may change the use settings of the select cartridge 100b based on the detergent information included in the NFC tag 160 of the mounted select cartridge 100b (845). For example, the washing machine 1 may detect a change in the detergent accommodated in the select cartridge 100b through the NFC tag 160 and the NFC reader 920, and automatically change the use settings of the select cartridge 100b based on the information about the new detergent of the select cartridge 100b. The use settings of the normal cartridge 100a select cartridge 100b may include, e.g., information about the detergent such as the type of detergent accommodated in the select cartridge 100b and the input amount of the detergent.

[0162] According to an embodiment, the washing machine 1 may store the current use settings of the changed select cartridge 100b in the memory 191 (850).

[0163] According to an embodiment, when the information about the detergent of the normal cartridge 100a is not changed (NO in 815), the washing machine 1 may perform the above-described operation 830.

[0164] Specifically, FIG. 9 is a view illustrating an automatic setting operation of a cartridge according to a change in contents accommodated in a select cartridge 100b. For example, FIG. 9 exemplarily illustrates an automatic setting operation of a detergent cartridge when a select cartridge 100b having different content is mounted on the detergent supply part 90 in a state in which the normal cartridge 100a in which the normal detergent is accommodated is mounted on the detergent supply part 90. FIG. 10 is a view exemplarily illustrating a user notification through an external electronic device 200 when the content of a select cartridge 100b is changed to a normal detergent.

[0165] Referring to FIGS. 9 and 10, when the select cartridge 100b in which the softener is accommodated is mounted on the detergent supply part 90, the washing machine 1 may obtain information about the detergent (e.g., softener) included in the NFC tag 160 of the select cartridge 100b through the NFC reader 920, and automatically change the use settings of the select cartridge 100b to correspond to the softener based on the obtained information. For example, the washing machine 1 may change the type of detergent used in the select cartridge 100b to softener, and change the input amount of detergent to a standard amount for each manufacturer of the accommodated softener.

[0166] When the select cartridge 100b in which the normal detergent is accommodated is mounted on the detergent supply part 90, the washing machine 1 may obtain information about the detergent (e.g., the normal detergent) included in the NFC tag 160 of the select cartridge 100b through the NFC reader 920, and automatically change the

use settings of the select cartridge 100b to correspond to the normal detergent based on the obtained information. For example, the washing machine 1 may change the type of detergent used in the select cartridge 100b to normal detergent, and change the input amount of detergent to a standard amount for each manufacturer of the accommodated normal detergent. Further, the washing machine 1 may provide an alarm regarding selection of a detergent container to be used for washing to the external electronic device 200. For example, the text 1010 “Normal detergent is in the normal cartridge and the select cartridge. Please choose the cartridge to be used for washing” and the text 1020 “normal cartridge” and the text 1030 “select cartridge” for selection by the user may be output on the display screen 1000 of the external electronic device 200 at the request of the washing machine 1. Further, the washing machine 1 may provide an alarm regarding whether to use a softener to the external electronic device 200. For example, the text 1040 “Softener is not currently in the cartridge. Please check whether to use the softener” may be output on the display screen 1000 of the external electronic device 200 at the request of the washing machine 1.

[0167] When the select cartridge 100b in which the neutral detergent is accommodated is mounted on the detergent supply part 90, the washing machine 1 may obtain information about the detergent (e.g., the neutral detergent) included in the NFC tag 160 of the select cartridge 100b through the NFC reader 920, and automatically change the use settings of the select cartridge 100b to correspond to the neutral detergent based on the obtained information. For example, the washing machine 1 may change the type of detergent used in the select cartridge 100b to neutral detergent, and change the input amount of detergent to a standard amount for each manufacturer of the accommodated neutral detergent. Further, the washing machine 1 may provide an alarm regarding whether to use a softener to the external electronic device 200, as described above.

[0168] When the select cartridge 100b in which the infant detergent is accommodated is mounted on the detergent supply part 90, the washing machine 1 may obtain information about the detergent (e.g., the infant detergent) included in the NFC tag 160 of the select cartridge 100b through the NFC reader 920, and automatically change the use settings of the select cartridge 100b to correspond to the infant detergent based on the obtained information. For example, the washing machine 1 may change the type of detergent used in the select cartridge 100b to infant detergent, and change the input amount of detergent to a standard amount for each manufacturer of the accommodated infant detergent. Further, the washing machine 1 may provide an alarm regarding whether to use a softener to the external electronic device 200, as described above.

[0169] FIG. 11 is a flowchart illustrating a method for controlling a washing machine according to an embodiment of the disclosure.

[0170] FIG. 12 is a view exemplarily illustrating an alarm operation of a washing machine per remaining amount of detergent accommodated in a detergent cartridge according to an embodiment of the disclosure.

[0171] Referring to FIGS. 11 and 12, the washing machine 1 according to an embodiment may measure the resistances of the electric lines 140a and 140b through the current sensor 183, and may estimate the remaining amount of detergent accommodated in the detergent cartridges 100,

e.g., the normal cartridge **100a** and the select cartridge **100b**, based on the measured resistances.

[0172] According to an embodiment, the washing machine **1** may receive and set an alarm for each remaining amount of detergent accommodated in the detergent cartridge **100** from the input device **51** of the control panel **50** or the external electronic device **200** (**1110**). For example, the user may set a plurality of alarms corresponding to the remaining amount of detergent for which the user wants to receive a notification through the input device **51** or the external electronic device **200**. As illustrated in FIG. **12**, the alarm settings for each remaining amount of detergent accommodated in the detergent cartridge **100** may include a first alarm provided to the user when the initial content, i.e., the remaining amount of detergent corresponding to 70% of the total amount of detergent, is detected, a second alarm provided to the user when the remaining amount of detergent corresponding to 40% of the total amount of detergent is detected, a third alarm provided to the user when the remaining amount of detergent corresponding to 20% of the total amount of detergent is detected, a fourth alarm provided to the user when the remaining amount of detergent corresponding to 10% of the total amount of detergent is detected, and an nth alarm provided to the user when the remaining amount of detergent corresponding to xx % of the total amount of detergent is detected.

[0173] According to an embodiment, the washing machine **1** may measure the current resistances of the electric lines **140a** and **140b**, which vary according to the discharge of the detergent, through the current sensor **183** (**1120**). For example, as described above with reference to FIGS. **6A** and **6B**, when the detergent is discharged, the partition plate **120** or the connector **150** moves in the horizontal direction (e.g., right), and the resistances of the electric lines **140a** and **140b** also change. Since the horizontal displacement of the partition plate **120** or the connector **150** according to the discharge of the detergent substantially corresponds to the remaining amount of the detergent, the current remaining amount of the detergent may be estimated based on the current resistances of the electric lines **140a** and **140b** measured through the current sensor **183**.

[0174] According to an embodiment, the washing machine **1** may determine whether the current remaining amount of the detergent meets the alarm setting set by the user (**1130**).

[0175] According to an embodiment, when the current remaining amount of detergent meets the alarm setting set by the user (YES in **1130**), the washing machine **1** may provide a notification of the current remaining amount of detergent to the user through the output device **52** or the external electronic device **200** (**1140**). For example, when the remaining amount of detergent estimated using the current resistance measured in operation **1120** corresponds to 70% of the total amount of detergent, the washing machine **1** may provide an alarm including information indicating that the current remaining amount of detergent is 70% to the user through the output device **52** or the external electronic device **200**.

[0176] According to an embodiment, when the current remaining amount of the detergent does not meet the alarm setting set by the user (NO in **1130**), the washing machine **1** may perform operation **1130** again, continuing to determine whether the current remaining amount of the detergent meets the alarm setting set by the user.

[0177] By such a series of operations, the washing machine **1** may monitor the remaining amount of detergent accommodated in the detergent cartridge **100** in real time, and provide an alarm for each remaining amount of detergent corresponding to the alarm setting set by the user to the user.

[0178] FIG. **13** is a perspective view illustrating a washing machine according to an embodiment of the disclosure.

[0179] FIG. **14** is a view illustrating a drawn-out state of a detergent supply device according to an embodiment of the disclosure.

[0180] FIG. **15** is an exploded perspective view illustrating a detergent supply device according to an embodiment of the disclosure.

[0181] Referring to FIGS. **13** to **15**, a washing machine **1a** according to an embodiment may include a cabinet **10** forming an outer appearance, a tub (e.g., the tub **20** of FIG. **2**) provided inside the cabinet **10** to store mixed water of water and a detergent, a drum (e.g., the drum **30** of FIG. **2**) provided inside the tub **20** to accommodate laundry and rotating, a door **40** rotatably coupled to the cabinet **10**, a control panel **50** receiving an operation command of the washing machine **1a** or outputting operation information, a driving device (e.g., the driving device **60** of FIG. **2**) rotating the drum **30**, a detergent supply part **90** for supplying the detergent to the tub **20**, and a detergent supply device **300** accommodating the detergent to supply the accommodated detergent to the tub (e.g., the tub **20** of FIG. **2**).

[0182] The washing machine **1a** may be substantially identical or similar in structure, shape, and/or operation to the washing machine **1** illustrated in FIGS. **1** and **2**. In this case, the description of the washing machine **1** made above with reference to FIGS. **1** and **2** may be substantially identically applied to the washing machine **1a**. Hereinafter, for ease of description, the same parts as those described above are skipped from the description.

[0183] According to an embodiment, the detergent supply device **300** may be provided to be drawn in or out from the cabinet **10**. For example, the detergent supply device **300** may be drawn into the cabinet **10** or drawn out of the cabinet **10** through the opening **90a** of the detergent supply part **90** provided in the front upper portion of the cabinet **10**. For example, the detergent supply device **300** may be provided in a drawer form as a whole.

[0184] When the detergent supply device **300** is drawn into the cabinet **10**, during the operation (e.g., the washing cycle) of the washing machine **1a**, the detergent accommodated in the detergent supply device **300** may be mixed with water supplied through a water supply pipe (e.g., the water supply pipe **14a** of FIG. **2**), and then may be supplied to the tub **20** along the supply pipe (e.g., the supply pipe **16** of FIG. **2**).

[0185] When the detergent supply device **300** is drawn out the cabinet **10**, the user may separate the detergent container (e.g., the first detergent container **320**) mounted on the detergent supply device **300** to be described below and replace it with a new detergent container **320**, refill the detergent container (e.g., the second detergent container **330**) with the same type of detergent, remove the existing detergent from the detergent container **330** and then change it to another detergent, or separate the detergent container **330** from the washing machine **1a** and wash it.

[0186] According to an embodiment, the detergent supply device **300** may include a housing **310** forming an outer

appearance, and a plurality of detergent containers **320** and **330** detachably coupled to the housing **310** and accommodating detergent and a detergent pump (not shown) fixed to the housing **310**.

[0187] According to an embodiment, the housing **310** may include a front cover **311** provided with a handle **311a** for drawing in/out the detergent supply device **300**, and a detergent container mounting portion **312** extending rearward of the front cover **311** and detachably coupled to the detergent containers **320** and **330**.

[0188] According to an embodiment, the front cover **311** of the housing **310** may be disposed on the same plane (e.g., front surface) as the front panel **11a** and/or the control panel **50** of the cabinet **10** in the drawn-in state of the detergent supply device **300**. However, the disclosure is not limited thereto.

[0189] According to an embodiment, the detergent container mounting portion **312** of the housing **310** may be partitioned by a plurality of partition walls, and may include a plurality of detergent container mounting portions **3121**, **3122**, **3123**, and **3124** having an open upper side. For example, the detergent container mounting portion **312** may include a 1-1th detergent container mounting portion **3121** where a 1-1th detergent container **320a** to be described below is mounted, a 1-2th detergent container mounting portion **3122** where a 1-2th detergent container **320b** to be described below is mounted, a 2-1th detergent container mounting portion **3123** where a 2-1th detergent container **330a** to be described below is mounted, and a 2-2th detergent container mounting portion **3124** where a 2-2th detergent container **330b** to be described below is mounted.

[0190] According to an embodiment, the plurality of detergent container mounting portions **3121**, **3122**, **3123**, and **3124** may be provided to correspond to the size, shape, and/or arrangement of the detergent containers **320a**, **320b**, **330a**, and **330b**.

[0191] In an embodiment, the central portion **3121c** of the 1-1th detergent container mounting portion **3121** may be designed to have different widths from the front portion **3121a** and the rear portion **3110b** of the 1-1th detergent container mounting portion **3121**. In an embodiment, the central portion **3122c** of the 1-2th detergent container mounting portion **3122** may be designed to have different widths from the front portion **3122a** and the rear portion **3122b** of the 1-2th detergent container mounting portion **3122**. In an embodiment, the 1-1th detergent container mounting portion **3121** and the 1-2th detergent container mounting portion **3122** may be provided to be symmetrical left and right. In an embodiment, the 2-1th detergent container mounting portion **3123** may be provided between the central portion **3121c** of the 1-1th detergent container mounting portion **3121** and the central portion **3122c** of the 1-2th detergent container mounting portion **3122**. In an embodiment, the 2-2th detergent container mounting portion **3124** may be provided between the central portion **3121c** of the 1-1th detergent container mounting unit **3121** and the central portion **3122c** of the 1-2th detergent container mounting portion **3122**. In an embodiment, the 2-1th detergent container mounting portion **3123** may be disposed in front of the 2-2th detergent container mounting portion **3124**.

[0192] According to an embodiment, the plurality of detergent containers **320** and **330** may be designed differ-

ently depending on the type of detergent accommodated in the detergent container and/or whether the detergent is automatically put in.

[0193] According to an embodiment, the plurality of detergent containers **320** and **330** may include a plurality of first detergent containers **320** in which a detergent for automatic input is accommodated.

[0194] According to an embodiment, the plurality of first detergent containers **320** may include a 1-1th detergent container **320a** and a 1-2th detergent container **320b** that are provided to selectively contact the detergent pump (not shown) to discharge the accommodated detergent in accordance with the use settings of the detergent container.

[0195] In an embodiment, the 1-1th detergent container **320a** may accommodate a detergent. In an embodiment, the 1-2th detergent container **320b** may accommodate a detergent or a softener. The first detergent container **320** may be referred to as an automatic detergent container, the 1-1th detergent container **320a** may be referred to as a normal detergent container, and the 1-2th detergent container **320b** may be referred to as a selective detergent container.

[0196] According to an embodiment, the 1-1th detergent container **320a** may be detachably mounted on the first detergent container mounting portion **3121**. According to an embodiment, the 1-2th detergent container **320b** may be detachably mounted on the second detergent container mounting portion **3122**. In an embodiment, the 1-1th detergent container **320a** and the 1-2th detergent container **320b** may be disposed left and right. The 1-2th detergent container **320b** may have a structure of being substantially horizontally symmetrical with the 1-1th detergent container **320a**.

[0197] According to an embodiment, the first detergent container **320** may include a detergent container identification member **327** disposed on an upper side of the main body **321** and including information about the detergent accommodated in the first detergent container **320**. For example, the detergent container identification member **327** may be implemented in the form of an NFC tag. Hereinafter, for convenience of description, the detergent container identification member **327** is referred to as an NFC tag **327**. The washing machine **1a** may obtain information about the detergent accommodated in the first detergent container **320** through the NFC tag **327** of the first detergent container **320**, and may automatically set the use settings of the first detergent container **320** based on the obtained information about the detergent. The description made with reference to FIGS. **8** to **10** may be applied to the automatic setting method of the detergent container **320** using the NFC tag **327**. For example, when the first detergent container **320** is drawn in the detergent supply part **90**, the washing machine **1a** may obtain information about the detergent included in the NFC tag **327** of the first detergent container **320** through the NFC reader (e.g., the NFC reader **920** of FIGS. **6A** and **6B**) provided in the detergent supply part **90**, and may automatically change the use settings of the first detergent container **320** based on the obtained information about the detergent.

[0198] According to an embodiment, the plurality of detergent containers **320** and **330** may include a plurality of second detergent containers **330** in which a detergent for manual input is accommodated.

[0199] According to an embodiment, the plurality of detergent containers **330** may include a 2-1th detergent container **330a** and a 2-2th detergent container **330b** in

which a detergent mixed with water supplied from the water supply pipe **14a** is accommodated when the washing machine is operated. In an embodiment, the 2-1th detergent container **330a** may be disposed in front of the 2-2th detergent container **330b** and may accommodate the detergent. In an embodiment, the 2-1th detergent container **330a** may accommodate a detergent. In an embodiment, the 2-2th detergent container **330b** may accommodate a softener. The second detergent container **330** may be referred to as a manual detergent/softener container, the 2-1th detergent container **330a** may be referred to as a manual detergent container, and the 2-2th detergent container **330b** may be referred to as a manual softener container.

[0200] According to an embodiment, the 2-1th detergent container **330a** may be detachably mounted on the 2-1th detergent container mounting portion **3123**, and may be disposed between the 1-1th detergent container **320a** and the 1-2th detergent container **320b**. In an embodiment, the 2-1th detergent container **330a** may have an open upper side and may have a cup shape as a whole.

[0201] According to an embodiment, the 2-2th detergent container **330b** may be detachably mounted on the 2-2th detergent container mounting portion **3124**, and may be disposed between the 1-1th detergent container **320a** and the 1-2th detergent container **320b**. In an embodiment, the 2-2th detergent container **330b** may be disposed behind the 2-1th detergent container **330a**. The 2-2th detergent container **330b** may have substantially the same structure as the 2-1th detergent container **330a**. For example, the 2-2th detergent container **330b** may have an open upper side and may have a cup shape as a whole.

[0202] As described above, the washing machine may automatically configure the use settings of the detergent cartridge (or automatic detergent container) through communication between the NFC tag **160** provided to the detergent cartridge (or automatic detergent container) and the NFC reader **920** of the detergent supply part **90**, eliminating the need for the user's action for the use settings and hence enhancing the user's convenience.

[0203] A washing machine **1** according to an embodiment of the disclosure may include a cabinet **10**. The washing machine **1** may include a tub **20** that is disposed in the cabinet **10** to contain water for washing. The washing machine **1** may include a detergent supply part **90** for supplying detergent to the tub **20**. The washing machine may include a detergent cartridge **100** storing the detergent, provided to be drawn in or out the detergent supply part **90**, and a cartridge identification member **160** including information about the detergent stored in the detergent cartridge **100**. The washing machine may include a cartridge identification device **920** positioned near the cartridge identification member **160** when the detergent cartridge **100** is drawn in the detergent supply part **90** to recognize the cartridge identification member.

[0204] According to an embodiment, the washing machine may include a controller **190** configured to obtain information about the detergent of the detergent cartridge **100** included in the cartridge identification member **160** through the cartridge identification device **920** and change a use setting of the detergent cartridge **100** based on the obtained information.

[0205] According to an embodiment, the use setting of the cartridge **100** may include a setting regarding an input

amount of the detergent accommodated in the cartridge **100** and a type of the detergent accommodated in the cartridge **100**. According to an embodiment, the information about the detergent accommodated in the cartridge **100** may include a type of the detergent and a standard input amount of the detergent. The standard input amount of the detergent may correspond to a detergent input amount according to a washing load.

[0207] According to an embodiment, the cartridge **100** may include a main body **110** including a first space **110a** storing air and a second space **110b** storing the detergent. The cartridge **100** may include a partition plate **120** disposed to be horizontally movable inside the main body **110** and partitioning the first space **110a** and the second space **110b**. The cartridge **100** may include a nozzle device **130** including a flow path **1311** selectively communicating with the second space **110b** to discharge the detergent to an outside.

[0208] The washing machine **1** may further comprise a nozzle pressurizer **930** configured to be horizontally movable and selectively contacting the nozzle device **130** to pressurize the nozzle device **130**. The nozzle device **130** may discharge the detergent to the outside as the flow path **1311** and the second space **110b** communicate with each other when pressurized according to a horizontal movement of the nozzle pressurizer **930**.

[0209] According to an embodiment, the main body **110** may include a pair of electric lines **140a**, **140a'**, **140b**, **140b'** extending from the first space **110** to the second space **110b** along an inner surface of the main body **110** and spaced apart from each other.

[0210] According to an embodiment, the detergent cartridge **100** may include a connector **150** disposed on one surface of the partition plate **120** facing the first space **110a** to electrically connect the pair of electric lines **140a**, **140a'**, **140b**, **140b'**.

[0211] According to an embodiment, the detergent supply part **90** may include a detergent supply part housing **910** defining an opening **90a** for drawing in/out the detergent cartridge **100** and a space **90b** for accommodating the detergent cartridge **100**. The washing machine **1** may further comprise a current sensor **183** disposed on one surface **912** of the detergent supply part housing **910** and electrically connected to the electric lines **140a**, **140a'**, **140b**, **140b'** of the detergent cartridge **100** when the detergent cartridge **100** is drawn in to measure a resistance of the electric lines **140a**, **140a'**, **140b**, **140b'**.

[0212] According to an embodiment, the washing machine **1** may further comprise a controller **190** configured to determine whether the detergent cartridge **100** is mounted in the detergent supply part **90** based on the resistance measured through the current sensor **183**.

[0213] According to an embodiment, the connector **150** may be moved in a direction away from the first space **110a** when the detergent accommodated in the second space **110b** is discharged. According to an embodiment, the washing machine **1** may further comprise a controller **190** configured to detect a change in the resistance of the pair of electric lines **140a**, **140a'**, **140b**, **140b'** according to a movement of the connector **150** through the current sensor **183** and measure a remaining amount of the detergent based on the detected change in the resistance.

[0214] According to an embodiment, the pair of electric lines **140a**, **140a'**, **140b**, **140b'** may be bent multiple times to have an irregular pattern or a wavy pattern.

[0215] A washing machine 1 according to an embodiment of the disclosure may include a cabinet 10. The washing machine 1 may include a tub 20 that is disposed in the cabinet 10 to contain water for washing. The washing machine 1 may include a detergent supply part 90 for supplying detergent to the tub 20. The washing machine 1 may comprise a detergent cartridge 100 storing the detergent and provided to be drawn in or out the detergent supply part 90. The detergent cartridge 100 may include a main body 110 including a first space 110a storing air and a second space 110b storing the detergent, a partition plate 120 disposed to be horizontally movable inside the main body 110 and partitioning the first space 110a and the second space 110b, a pair of electric lines 140a, 140a', 140b, 140b' extending from the first space 110 to the second space 110b along an inner surface of the main body 110 and spaced apart from each other, and a connector 150 disposed on one surface of the partition plate 120 facing the first space 110a to electrically connect the pair of electric lines 140a, 140a', 140b, 140b'.

[0216] According to an embodiment, the detergent supply part 90 may include a detergent supply part housing 910 defining an opening 90a for drawing in/out the detergent cartridge 100 and a space 90b for accommodating the detergent cartridge 100. The washing machine 1 may further comprise a current sensor 183 disposed on one surface 912 of the detergent supply part housing 910 and electrically connected to the electric lines 140a, 140a', 140b, 140b' of the detergent cartridge 100 when the detergent cartridge 100 is drawn in to measure a resistance of the electric lines 140a, 140a', 140b, 140b'.

[0217] According to an embodiment, the washing machine 1 may further comprise a controller 190 configured to determine whether the detergent cartridge 100 is mounted in the detergent supply part 90 based on the resistance measured through the current sensor 183.

[0218] According to an embodiment, the connector 150 may be moved in a direction away from the first space 110a when the detergent accommodated in the second space 110b is discharged. According to an embodiment, the washing machine 1 may further comprise a controller 190 configured to detect a change in the resistance of the pair of electric lines 140a, 140a', 140b, 140b' according to a movement of the connector 150 through the current sensor 183 and measure a remaining amount of the detergent based on the detected change in the resistance.

[0219] According to an embodiment, the pair of electric lines 140a, 140a', 140b, 140b' may be bent multiple times to have an irregular pattern or a wavy pattern.

[0220] A method for controlling a washing machine including a cabinet 10, a tub 20 disposed in the cabinet 10 to contain water for washing, a detergent supply part 90 for supplying a detergent to the tub 20, and a detergent cartridge 100 storing the detergent therein and detachably coupled to the detergent supply part 90, according to an embodiment of the disclosure, may comprise recognizing a detergent container identification member 160 through a detergent container identification device 920 when the detergent cartridge 100 is drawn in the detergent supply part 90 and obtaining information about the detergent of the detergent cartridge 100 included in the detergent container identification member 160. The method for controlling the washing machine 1 may comprise changing a use setting of the detergent

cartridge 100 based on the obtained information about the detergent of the detergent cartridge 100.

[0221] According to an embodiment, the use setting of the detergent cartridge 100 may include a setting regarding an input amount of the detergent accommodated in the detergent cartridge 100 and a type of the detergent accommodated in the detergent cartridge 100.

[0222] According to an embodiment, the method for controlling the washing machine 1 may further comprise measuring a resistance of a pair of electric lines 140a, 140a', 140b, 140b' provided in the detergent cartridge 100 through a current sensor 183 and determining whether the detergent cartridge 100 is mounted in the detergent supply part 90 based on the measured resistance.

What is claimed is:

1. A washing machine, comprising:

- a cabinet;
 - a tub in the cabinet;
 - a detergent supply part for supplying a detergent to the tub;
 - a detergent cartridge accommodating the detergent, configured to be drawn in or out the detergent supply part, and including a cartridge identification member storing information about the detergent accommodated in the detergent cartridge; and
 - a cartridge identification device positioned to, with the detergent cartridge drawn into the detergent supply part, recognize the cartridge identification member.
2. The washing machine of claim 1, further comprising:
- a controller configured, with the detergent cartridge drawn into the detergent supply part, to obtain the information about the detergent accommodated in the detergent cartridge stored in the cartridge identification member through the cartridge identification device, and change a use setting of the detergent cartridge based on the obtained information.

3. The washing machine of claim 2, wherein the use setting of the detergent cartridge includes a setting regarding an input amount of the detergent accommodated in the detergent cartridge and a type of the detergent accommodated in the detergent cartridge.

4. The washing machine of claim 1, wherein the information about the detergent accommodated in the detergent cartridge includes a type of the detergent and a standard input amount of the detergent, and the standard input amount of the detergent corresponds to a detergent input amount according to a washing load.

5. The washing machine of claim 1,

wherein the detergent cartridge includes:

- a main body including a first space configured to accommodate air and a second space configured to accommodate the detergent,
- a partition plate horizontally movable inside the main body and partitioning the first space and the second space, and
- a nozzle device including a flow path selectively communicating with the second space to discharge the detergent to an outside of the detergent cartridge.

6. The washing machine of claim 5, further comprising:

- a nozzle pressurizer configured to be horizontally movable to selectively contact the nozzle device to pressurize the nozzle device,

wherein the nozzle device is configured to discharge the detergent to the outside of the detergent cartridge as the

flow path and the second space communicate with each other when pressurized according to a horizontal movement of the nozzle pressurizer.

7. The washing machine of claim 5, wherein the main body includes a pair of electric lines extending from the first space to the second space along an inner surface of the main body and spaced apart from each other.

8. The washing machine of claim 7, wherein the detergent cartridge includes a connector on a side of the partition plate facing the first space and electrically connecting the pair of electric lines.

9. The washing machine of claim 8, wherein the detergent supply part includes a detergent supply part housing including an opening through which the detergent cartridge is drawable into and drawable out from the detergent supply part, and a space to accommodate the detergent cartridge, and

the washing machine further includes:

a current sensor on a surface of the detergent supply part housing and that electrically connects to the electric lines of the detergent cartridge when the detergent cartridge is drawn into the detergent supply part and accommodated in the space of the detergent supply part, to measure a resistance of the electric lines.

10. The washing machine of claim 9, further comprising: a controller configured to determine whether the detergent cartridge is drawn into the detergent supply part based on the resistance measured through the current sensor.

11. The washing machine of claim 9, wherein the connector moves in a direction away from the first space when the detergent accommodated in the second space is discharged, and

the washing machine further comprises:

a controller configured to detect a change in the resistance of the electric lines according to a movement of the connector through the current sensor and measure a remaining amount of the detergent based on the detected change in the resistance.

12. The washing machine of claim 7, wherein the pair of electric lines are bent multiple times so as to have an irregular pattern or a wavy pattern.

13. A washing machine, comprising:

a cabinet;

a tub in the cabinet;

a detergent supply part for supplying a detergent to the tub; and

a detergent cartridge accommodating the detergent and configured to be drawn in or out the detergent supply part,

wherein the detergent cartridge includes:

a main body including a first space configured to accommodate air and a second space configured to accommodate a detergent,

a partition plate horizontally movable inside the main body and partitioning the first space and the second space,

a pair of electric lines extending from the first space to the second space along an inner surface of the main body and spaced apart from each other, and

a connector on a side of the partition plate facing the first space and electrically connecting the pair of electric lines; and

with the detergent cartridge drawn into the detergent supply part, the detergent accommodated in the second space is suppliable to the tub.

14. The washing machine of claim 13, wherein

the detergent supply part includes a detergent supply part housing including an opening through which the detergent cartridge is drawable into and drawable out from the detergent supply part, and a space to accommodate the detergent cartridge, and

the washing machine further comprises:

a current sensor on a surface of the detergent supply part housing and that electrically connects to the electric lines of the detergent cartridge when the detergent cartridge is drawn into the detergent supply part and accommodated in the space of the detergent supply part, to measure a resistance of the electric lines.

15. The washing machine of claim 14, further comprising: a controller configured to determine whether the detergent cartridge is drawn into the detergent supply part based on the resistance measured through the current sensor.

16. The washing machine of claim 14, wherein

the connector moves in a direction away from the first space when the detergent accommodated in the second space is discharged, and

the washing machine further comprises:

a controller configured to detect a change in the resistance of the electric lines according to a movement of the connector through the current sensor and measure a remaining amount of the detergent based on the detected change in the resistance.

17. The washing machine of claim 13, wherein the pair of electric lines are bent multiple times so as to have an irregular pattern or a wavy pattern.

18. A method for controlling a washing machine including a tub, a detergent supply part for supplying a detergent to the tub, a detergent cartridge accommodating the detergent therein, configured to be drawn in or out the detergent supply part, and including a cartridge identification member storing information about the detergent accommodated in the detergent cartridge, and a cartridge identification device positioned to, with the detergent cartridge drawn into the detergent supply part, recognize the cartridge identification member, the method comprising:

with the detergent cartridge drawn into the detergent supply part, recognizing the cartridge identification member through the cartridge identification device;

obtaining the information about the detergent accommodated in the detergent cartridge stored in the cartridge identification member from the recognized cartridge identification member; and

changing a use setting of the detergent cartridge based on the obtained information about the detergent accommodated in the detergent cartridge.

19. The method of claim 18, wherein the use setting of the detergent cartridge includes a setting regarding an input amount of the detergent accommodated in the detergent cartridge and a type of the detergent accommodated in the detergent cartridge.

20. The method of claim 18, wherein the detergent cartridge includes a first space configured to accommodate air, a second space configured to accommodate the detergent, and a pair of electric lines extending from the first

space to the second space and spaced apart from each other,
and the method further comprising:

measuring a resistance of the electric lines; and
determining whether the detergent cartridge is drawn into
the detergent supply part based on the measured resistance.

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